Online Appendix: Not For Publication

A Additional Material: Treatment Letters, Mail-In Survey, Project Website and the FEC Online Search Tool

Further Details on the Treatment Letters

Appendices A.1-A.5 show samples of the letters for different treatment types and sub-types. All these letter types shared basic characteristics. They all included the same header ("Boston, April 25th 2012") and the same last paragraph: "This letter is part of a study of political campaign contributions made by individuals which is being conducted by researchers at Harvard University. You can find more information about this project, including contact information, on our website." The letters included the web address of the project's website, shown in Appendix A.8, which provided basic information about the research project, and contact information to reach the research team and the University's Institutional Review Board. The main purpose of the website was to provide contextual information about our study to interested subjects, and to dissipate any doubts about its legitimacy, emphasizing its academic and non-partisan nature. Although the website provided some general information about the main research objective, to avoid the contamination of the experimental results, the website did not provide any details about the precise hypotheses to be tested, nor about the existence of several different treatment types. We directed individuals who were interested in receiving a debriefing brochure (a non-technical summary of the study's main hypotheses and results) to send an email to a dedicated address. We sent the brochure only after the data collection process was completed.

The mailing consisted of a single sheet of paper that folded and sealed to make a letter-size mailpiece. The outside of the mailpiece, a sample of which is shown in Appendix A.6, was the same for all treatment types. The design reflected two objectives. First, we wanted to maximize the credibility of the content. The outside of the mailpiece had the non-profit postage as well as the sender's Harvard address, in order to increase the recipient's confidence in the origin of the letter. We also wanted to maximize the recipient's interest in the letter and avoid it being discarded as junk mail. For this reason, we included a personalized message on the front (smaller font) and on the back (larger font) of the outside of the mailpiece. This message included the name of the recipient and indicated that the letter contained information about campaign contributions. Since all recipients had made contributions in the past, a personalized letter referring to this topic should have piqued the recipient's interest. However, in the middle of the election cycle these contributors probably received a great deal of unsolicited mail related to the campaign, so we expected that a majority of our letters would be discarded without even being opened. The implications for our estimates are discussed in Section 4.1 in the body of the paper.

A.1 Sample Letter: Website-Self

Boston, April 25th 2012

Dear John,

This letter is part of an effort to disseminate information about political campaign contributions made by individuals from your neighborhood:

Name of contributor	Amount - Party contributed to
S., ANITA	$600 – DEM
DOE, JOHN	**$375 – DEM**
T., WILLIAM JR	$1,000 – REP

Name of contributor	Amount - Party contributed to
H., ROBERT L	$300 – DEM
L., EDMOND	$2,500 – REP
G., LISA	$1,000 – REP

YOUR HOUSEHOLD WAS THE ONLY HOUSEHOLD RANDOMLY CHOSEN FROM YOUR AREA TO RECEIVE A LETTER OF THIS TYPE

The above table contains a list of the total campaign contributions to presidential candidates made by 6 individuals from your neighborhood in the period from April 1, 2011 to April 1, 2012, according to the public records published by the Federal Election Commission.

Your full name, address and details about your campaign contributions are freely available to anyone with Internet access. You can search for individual contributions by first and last name, or by zip code, using the following tool from the website of the Federal Election Commission:

www.fec.gov/finance/disclosure/norindsea.shtml

You can use this website to see which candidates or political parties your neighbors, friends, family and co-workers are contributing to. Access to the data is anonymous.

This letter is part of a study of political campaign contributions made by individuals which is being conducted by researchers at Harvard University. We will not send any more letters about past or future contributions to your household or to your neighbors. You can find more information about this project, including contact information, on our website:

Information Dissemination on Campaign Contributions

www.campaign-information.info

A.2 Sample Letter: Website-Neighbors

Boston, April 25th 2012

Dear John,

This letter is part of an effort to disseminate information about political campaign contributions made by individuals from your neighborhood:

Name of contributor	Amount - Party contributed to
S., ANITA	$600 – DEM
DOE, JANE	**$375 – DEM**
T., WILLIAM JR	$1,000 – REP

Name of contributor	Amount - Party contributed to
H., ROBERT L	$300 – DEM
L., EDMOND	$2,500 – REP
G., LISA	$1,000 – REP

YOUR HOUSEHOLD AND OTHER HOUSEHOLDS IN YOUR AREA WERE RANDOMLY CHOSEN TO RECEIVE A LETTER OF THIS TYPE

The above table contains a list of the total campaign contributions to presidential candidates made by 6 individuals from your neighborhood in the period from April 1, 2011 to April 1, 2012, according to the public records published by the Federal Election Commission.

Your full name, address and details about your campaign contributions are freely available to anyone with Internet access. You can search for individual contributions by first and last name, or by zip code, using the following tool from the website of the Federal Election Commission:

www.fec.gov/finance/disclosure/norindsea.shtml

You can use this website to see which candidates or political parties your neighbors, friends, family and co-workers are contributing to. Access to the data is anonymous.

This letter is part of a study of political campaign contributions made by individuals which is being conducted by researchers at Harvard University. We will not send any more letters about past or future contributions to your household or to your neighbors. You can find more information about this project, including contact information, on our website:

Information Dissemination on Campaign Contributions

www.campaign-information.info

A.3 Sample Letter: List-Once

Boston, April 25th 2012

Dear John,

This letter is part of an effort to disseminate information about political campaign contributions made by individuals. According to the public records of the Federal Election Commission, **this is a list of the political campaign contributions to presidential candidates made by 10 individuals from your neighborhood**:

Last name initial and first name of contributor	April 1, 2011 to April 1, 2012	
	Amount contributed	Party contributed to
DOE, JOHN	**$250**	**REP**
M., CHARLES	$1,000	DEM
C., SUSAN	$500	DEM
D., ANN	$500	DEM
B., CAROL	$250	DEM
L., ANNE	$212	DEM
W., CHARLOTTE T.	$200	DEM
W., MELANIE	$2,500	REP
P., JAMES	$2,000	REP
H., PATRICK	$750	REP

THIS IS THE ONLY TIME WE WILL CONTACT YOU. WE WILL NOT SEND A LIST OF FUTURE CONTRIBUTIONS TO YOUR HOUSEHOLD OR TO OTHER HOUSEHOLDS IN YOUR AREA.

This letter is part of a study of political campaign contributions made by individuals which is being conducted by researchers at Harvard University. You can find more information about this project, including contact information, on our website:

Information Dissemination on Campaign Contributions

www.campaign-information.info

A.4 Sample Letter: List-Update

Boston, April 25th 2012

Dear Jane,

This letter is part of an effort to disseminate information about political campaign contributions made by individuals. According to the public records of the Federal Election Commission, **this is a list of the political campaign contributions to presidential candidates made by 10 individuals from your neighborhood**:

Last name initial and first name of contributor	Past contributions: April 1, 2011 to April 1, 2012		*Future contributions to be reported:* May 1, 2012 to December 1, 2012	
	Amount contributed	Party contributed to	Amount contributed	Party contributed to
DOE, JANE	**$495**	**DEM**		
F., BEATRICE	$1,000	DEM		
W., FREDRICA	$712	DEM		
L., ANN	$250	DEM		
D., GENIEVE	$2,500	REP		
V., PAUL	$2,500	REP		
D., KATHERINE	$2,500	REP		
W., SETH	$2,500	REP		
R., ERIC	$1,000	REP		
E., STUART	$500	REP		

WE WILL SEND AN UPDATED LIST OF FUTURE CONTRIBUTIONS TO SOME HOUSEHOLDS. SOME OF YOUR NEIGHBORS MAY RECEIVE SUCH A LIST.

This letter is part of a study of political campaign contributions made by individuals which is being conducted by researchers at Harvard University. You can find more information about this project, including contact information, on our website:

Information Dissemination on Campaign Contributions

www.campaign-information.info

A.5 Sample Letter: List-Placebo

Boston, April 25th 2012

Dear Jane,

This letter is part of an effort to disseminate information about political campaign contributions made by individuals. Your household was randomly chosen to receive this information.

According to the Federal Election Commission, the limits on campaign contributions for individuals are as follows:

- $2,500 to each candidate or candidate committee per election.
- $30,800 to national party committee per calendar year.
- $10,000 to state, district & local party committee per calendar year.
- $5,000 to any other political committee per calendar year.
- $117,000 overall biennial limit ($46,200 to all candidates and $70,800 to all PACs and parties).

This letter is part of a study of political campaign contributions made by individuals which is being conducted by researchers at Harvard University. You can find more information about this project, including contact information, on our website:

Information Dissemination on Campaign Contributions

www.campaign-information.info

A.6 Outside of the Mailing

Outside of the mailing - front

DEAR JOHN:

IMPORTANT INFORMATION ABOUT POLITICAL CAMPAIGN CONTRIBUTIONS

Harvard University
108 Littauer Center
Cambridge, MA 02138

Nonprofit Org.
US Postage
PAID
Permit No. 492
04104

36

002*01**********3-DIGIT 040

TO: JOHN DOE
123 MAIN ST
SPRINGFIELD, VA 22150-1234

Outside of the mailing - back

DEAR JOHN:

IMPORTANT INFORMATION ABOUT POLITICAL CAMPAIGN CONTRIBUTIONS

Survey Instrument

36
002*01**********3-DIGIT 040
JOHN DOE
123 MAIN ST
SPRINGFIELD, VA 22150-1234

Cambridge, November 10th 2012

Dear John Doe,

We are researchers from Harvard University who are carrying out a non-partisan study about campaign contributions in the United States. This study includes a short survey designed to find out how much citizens know about the rules and regulations applying to individual campaign contributions.

We need your help for this study. We kindly ask you to take two minutes of your time to fill out this short, confidential survey and send it back to us in the pre-paid and pre-addressed envelope enclosed with this letter. You do not need to affix a stamp to the envelope. You can simply drop the letter into any US Postal Service mailbox. Of course, your participation is completely voluntary.

As a token of our gratitude for helping us with our research, all of those responding to this survey and mailing it back to us will automatically enter a lottery for 10 prizes of $100. Your chances in the lottery do not depend on your responses to the questions in the survey – Winners will be randomly chosen among all of those sending the survey back to us.

In the other side of this page you can find more information regarding the purpose of our research and the lottery prizes. You may keep this letter for your records – You do not need to send it back to us.

Sincerely,

Ricardo Perez-Truglia and Guillermo Cruces
The research team

Email: rtruglia@fas.harvard.edu
Address: Littauer Center G16R, 1805 Cambridge Street, Cambridge, Massachusetts 02138

Purpose of the study

This is part of a strictly academic project, and our research is not affiliated with any candidate or political party. The survey includes questions about certain features of the campaign contribution laws and regulations. The purpose of our research project is to study the implications of those features. Your responses to this survey will be confidential – Your responses will not be shared with anyone under any circumstances. Our research project has no commercial or political objective and is in compliance with the rules regulating the use of contribution information. If you are interested in receiving information about the results of the studies we are conducting, just send an email to rtruglia@fas.harvard.edu with the subject line "Debriefing" and we will send you information about our work as soon as our studies are finished.

Terms of the lottery

All the individuals who respond to the survey before January 31st 2013 (returned letters postmarked by that date) will be included in the lottery for the Amazon gift cards. Your odds of winning a prize will depend on the number of individuals who respond to the survey. For example, if - as expected - 1,500 individuals respond the survey, your chances of winning a prize will be of 1 in 150. Each individual can win a maximum of one (1) prize. If you are one of the winners, you will be notified by February 2013 by a letter sent to the same address where you received the survey. If you want us to use a different address, please state so in the space provided in the survey page.

For more information about this project, please visit the project's website

www.people.fas.harvard.edu/~rtruglia/campaign-survey.htm

Thank you in advance for your cooperation!

We are researchers from Harvard University who are carrying out a non-partisan study about campaign contributions in the United States. This is the short survey we mention in the accompanying letter. We kindly ask you to take two minutes of your time to fill out this short, confidential survey and send it back to us in the pre-paid and pre-addressed envelope enclosed with this letter. You do not need to affix a stamp to the envelope. You can simply drop the letter into any US Postal Service mailbox. Thank you for your cooperation.

1- Please tell us your gender

☐ Male ☐ Female

2- Your age (Please mark ONE option)

☐ Under 20 ☐ 20-29 ☐ 30-39 ☐ Over 40

3- What is the maximum contribution that an individual can legally make to a campaign committee per election? (Please mark ONE option)

☐ $1,500 ☐ $2,500 ☐ $4,000 ☐ I don't know

4- How accessible do you think that information on individuals' contributions to political campaigns is? If you do not know, please just give us your best guess. (Mark ONE option)

☐ The law says that the identity of contributors to political campaigns, the amounts contributed and the identity of recipients are all **public** information, and that information can be easily obtained by anyone with Internet access

☐ The law says that the identity of contributors to political campaigns, the amounts contributed and the identity of recipients are all **confidential** information which cannot be accessed by third parties.

5- How sure are you about your answer to the previous question? (Mark ONE option)

☐ Very sure ☐ Somewhat sure ☐ Unsure

6- In your answers to the previous two questions, you told us what YOU believe. Now we want to know what you think that OTHERS believe. Think about people in your neighborhood. What do you think their best guess would be about how accessible the information on individual campaign contributions is? (Mark ONE option)

- ☐ Most people would say that the identity of contributors to political campaigns, the amounts contributed and the identity of recipients are all **public** information.
- ☐ Most people would say that the identity of contributors to political campaigns, the amounts contributed and the identity of recipients are all **confidential** information.

7- Would you say that… (Mark ONE option)

- ☐ A **vast majority** of your neighbors believe that information on individual campaign contributions is **public.**
- ☐ A **majority** of your neighbors believe that information on individual campaign contributions is **public.**
- ☐ A **majority** of your neighbors believe that information on individual campaign contributions is **confidential.**
- ☐ A **vast majority** of your neighbors believe that information on individual campaign contributions is **confidential.**

8- In your opinion, how much do you think a politically engaged individual with an average income should contribute to a presidential campaign per election cycle (every four years)?

Please enter an amount: $ _____

Respondent: John Doe (123 MAIN ST, SPRINGFIELD, VA 22150-1234)

Thanks for your response! We will contact you by mail if you win one of the lottery prizes. If needed, please write down an alternative address to receive any further correspondence about the prize:

- -

- -

A.8 Text Displayed on the Project's Website Mentioned in the Letters

Welcome to our website. We are a group of researchers at Harvard University studying political campaign contributions made by individuals. With that goal, we are sending out personalized mailings about campaign contributions in the U.S. If you received a letter and have any questions about the information provided to you, or our research, please feel free to email us at link and we will get back to you as soon as possible.

The purpose of our research project is to study the implications of the public's awareness about the open nature of campaign contributions. The ultimate goal is to understand the different mechanisms through which the open nature of this information may affect contributions. We hope that the research will shed light on the advantages and disadvantages of alternative disclosure policies, which we believe is a very important issue. If you are interested in receiving information about the results of the studies we are conducting, just send us a blank email to link and we will send information about our work as soon as our studies are finished. This is part of a strictly academic project, and our research is not affiliated with any candidate or political party. All the information that we used in our mailings is publicly accessible through the website of the Federal Election Commission (FEC). This website includes a search tool with which anyone can access information about individual contributions by donor name (link). This research team at Harvard includes Ricardo Perez-Truglia, a PhD student in Economics (link), and Dr. Guillermo Cruces (link). Mr. Perez-Truglia's primary thesis advisor is Professor Nadarajan Chetty. You may write to Ricardo Perez-Truglia directly at the above address and you may also reach his faculty thesis advisers by writing to link.

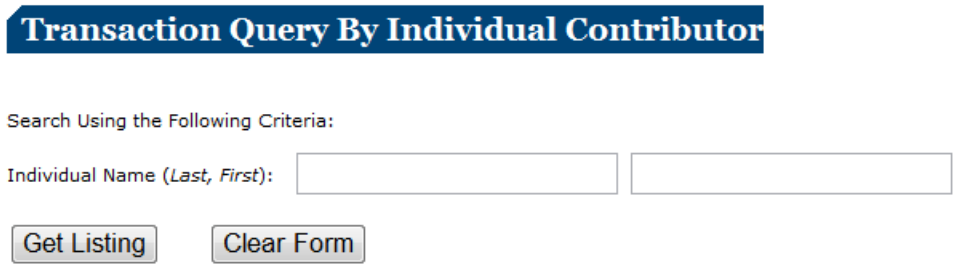
All individuals who received a letter about campaign donations were randomly selected by an automated computer program from the public records of the FEC. The information provided in the letter was available from public records and was selected without regard to party affiliation. The FEC explicitly allows the use of information about individual campaign contributions for academic research such as this project. The specific activities identified as permissible by the Federal Election Commission include the use of individual contributor information for bona fide academic research projects that do not involve the sale or use of that information for a commercial purpose or for soliciting contributions (see FEC Advisory Opinion No. 1986-25). Our research project has no commercial or political objective and is in compliance with the rules regulating the use of contribution information. For more information, please see the FEC's "sale and use brochure" (link). This project was reviewed and approved in advance by Professor Chetty and by the Committee on the Use of Human Subjects in Research, a research ethics committee (also known as an "institutional review board" or "IRB") at Harvard University. Complaints or problems concerning any research project may, and should, be reported if they arise. The Committee can be reached via email (link) or by telephone (link).

Thank you again for your visit to this website and for your interest in our research.

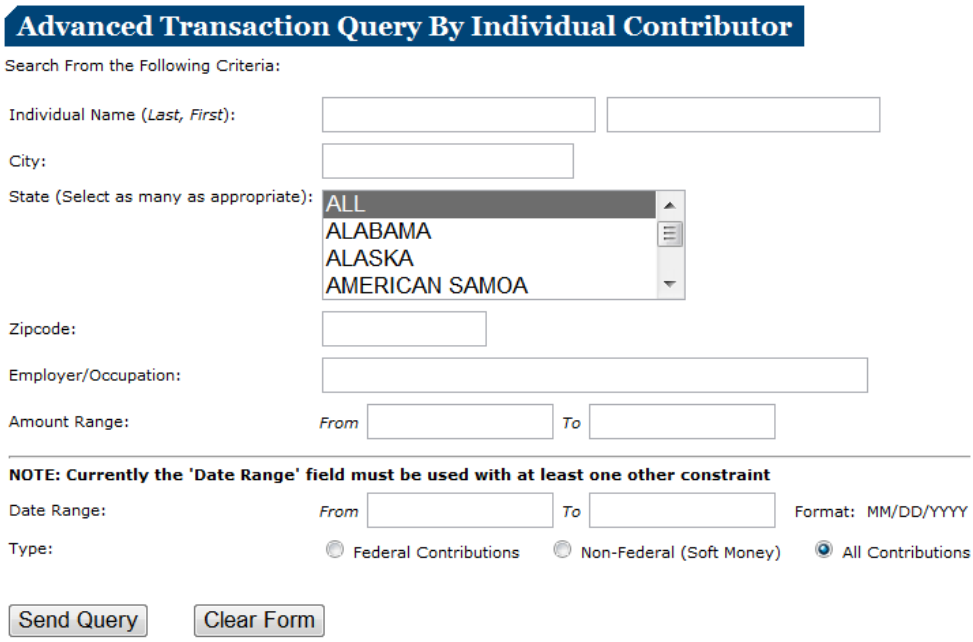
Ricardo Perez-Truglia and Guillermo Cruces (the research team)

A.9 Snapshots of the FEC Website's Search Tool

The FEC provides an easily accessible online database of individual campaign contributions. The database can searched by first and/or last name:



Advanced search can be done by other criteria, such as city, state, date range, and so forth:



This is a sample of how the search results are displayed (they are the same for basic and advanced search). This sample is for one transaction - the search tool displays one record per transaction:

Individual Contributions Arranged By Type, Giver, Then Recipient

Contributions to Political Committees

DOE, JOHN
ELIOT, ME 03903
HOMEMAKER

PAUL, RON
VIA RON PAUL 2012 PRESIDENTIAL CAMPAIGN COMMITTEE INC.

12/16/2011 250.00 12345678900

The (fictitious) number 12345678900 has a hyperlink to the exact page of the Schedule A-P corresponding to the transaction. The following is a sample Schedule A-P:

SCHEDULE A-P
ITEMIZED RECEIPTS

Use separate schedule(s) for each category of the Detailed Summary Page

FOR LINE NUMBER: (check only one)
☐ 16 ☒ 17a ☐ 17b ☐ 17c ☐ 17d ☐ 18
☐ 19a ☐ 19b ☐ 20a ☐ 20b ☐ 20c ☐ 21

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Any information copied from such Reports and Statements may not be sold or used by any person for the purpose of soliciting contributions or for commercial purposes, other than using the name and address of any political committee to solicit contributions from such committee.

NAME OF COMMITTEE (In Full)
Ron Paul 2012 Presidential Campaign Committee Inc.

A. Full Name (Last, First, Middle Initial)
Doe, John

Mailing Address 123 Fake Street

City: Eliot
State: ME
Zip Code: 03903

FEC ID number of contributing federal political committee. C

Name of Employer: Homemaker
Occupation: Homemaker

Receipt For: 2012
☒ Primary ☐ General
☐ Other (specify) ▼

Election Cycle-to-Date ▼ 500.00

Transaction ID : 0992651

Date of Receipt
M M / D D / Y Y Y Y
12 / 16 / 2011

Amount of Each Receipt this Period
250.00

B Further Details on the Subject Pool and on the Implementation of the Field Experiment

B.1 Subject Pool and Timing of the Mailing

As detailed in Section 3, a total of 280,456 unique individuals were listed in the FEC records as having made a contribution to a presidential candidate between April 1, 2011 and April 1, 2012. This sample was obtained from the FEC's public records as of April 25, 2012, which includes contributions made until April 1 of that year. This sample of contributors excludes individuals contributing $200 or less over the course of the election cycle, as these individuals are not required to be reported to the FEC. While campaigns have increasingly relied on these donors (they represented 41.2% of all individual contributions in 2008 and 47.7% in 2012), the available evidence indicates that, besides the evident differences in income, those making small and large contributions are fairly similar.[i]

We discarded a substantial fraction of the original 280,456 contributors for data quality and other reasons, resulting in a final subject pool of 191,832 individuals. We present here a list of the most important reasons and criteria. We do not report what percentage of individuals were excluded for each reason because a majority of the excluded individuals were excluded for multiple reasons. We excluded observations for which the address information was invalid and could not be corrected (e.g., missing street number). We also excluded individuals reporting addresses used by more than two unique individuals (which most likely corresponds to work addresses) and individuals who provided P.O. boxes as their home address. We matched the address information to the NCOA database to identify individuals or households that changed residence over the previous 18 months, and we excluded all individuals who changed residence since the date when they made their first contribution during the election cycle. We excluded individuals who reported contributions in multiple addresses. We excluded individuals whose mean distance (as the crow flies) from the ten closest contributors was over three miles. We also excluded individuals who had already made a total contribution over $1,500, all contributors living outside the 50 U.S. States, and all contributors in Washington D.C.. Finally, we also discarded individuals who had simultaneously made contributions over $200 to the Obama campaign and to at least one of the Republican presidential candidates – only 0.08% of the original sample belonged in this category.

We mailed the letters depicted in the previous paragraph on May 6, 2012. The date of delivery of each letter is an important factor to consider when determining exposure to our information treatment. We were able to track the delivery status of each letter through the USPS scanning

[i] McAdams, J. and Green, J.C. (2002). "Fat Cats and Thin Kittens: Are People Who Make Large Campaign Contributions Different?", CATO Institute Briefing Paper no. 76.

system, which does not confirm delivery but provides an estimate of when the letter was out for delivery (i.e., it tracks when and where each letter was last scanned). We generated a proxy for time of delivery equal to the most recent date when the letter was scanned if it was not forwarded or returned. For letters with incomplete tracking information, we imputed delivery information from other mail pieces in our batch delivered in the same 9-digit ZIP code. While the USPS tracking data is not a perfect indicator of delivery, it is a good approximation that provides a conservative lower bound for the actual date of delivery. Again, this proxy of delivery does not necessarily indicate that the letters were received or read, as the mailing did not include delivery confirmation service. We also constructed a proxy for the time when individuals may have read the letters. The letters included a link to a website with contact information for the research team and details about the research project. The website records indicate the number and date of visits. It is likely that individuals visited the website on the same day that they read the letter, or at least within the next few days. The distribution of visits to the website over time thus provides a proxy for the time when the individuals read the letters.

Figure B.1 compares our proxy of delivery date from the USPS tracking data with data on visits to the project's website. Figure B.1.a indicates that the number of letters in each State was almost exactly proportional to the number of unique visitors to the website – the R-squared for the regression line in the Figure is 0.98. This strong correlation indicates that the proxies for letter delivery and letters read are consistent. Figure B.1.b shows the distribution of new visitors to the website over time and the USPS-based proxy for mail delivery. The two distributions are very similar, although visits to the website seem to have a lag with respect to the proxy for delivery date. This is consistent with the fact that individuals do not necessarily read the mail the same day they get it. The difference in the right tail of the two distributions indicates that visits to the website sometimes occurred weeks after the letters were delivered. This probably corresponds to individuals who accumulate mail over time, or to those who were absent from their homes for some time. All in all, the evidence is consistent with our proxy for delivery being a conservative lower bound estimate of the actual date of delivery.

B.2 Descriptive Statistics, Subjects' Contribution Patterns and Balance Test for Randomization

Table B.1 presents summary statistics of individual characteristics from our experimental sample of early contributors (first column) compared to all contributors to presidential campaigns from the 2012 election cycle (second column) and to the general U.S. population (third column). The comparison between the first two columns indicates that the average contributor in our sample was fairly representative of all contributors to the 2012 presidential election to the extent that they

exhibit similar socio-economic characteristics, including racial composition and income. There are, however, some differences in contribution patterns between the two groups. Our subject pool contains a lower share of contributors to the Obama campaign. This is due to the fact that our subjects were early contributors and, because of the Republican primary, Republican candidates started their campaigns earlier. Our subject pool also has higher average contributions, which is partly due to the fact that Republicans, who contributed higher amounts, are over-represented in the experimental sample. Finally, the comparison of the first two columns with the third column illustrates the well documented fact that contributors are significantly different from the average U.S. citizen in several ways: e.g., contributors are more likely to be males, white and more likely to live in urban and wealthier areas.

Table B.2, in turn, presents summary statistics for a number of pre-treatment characteristics for each of the treatment types, including the amount of pre-treatment contributions and the party contributed to. As expected due to random assignment, the treatment groups are balanced in their observable characteristics. The last column reports the p-values from a test where the null hypothesis is that the means of the row variable for the six groups are equal. These tests indicate that the differences across treatments are not only very small but also not statistically significant.

Finally, Table B.3 details the pre and post-treatment contribution patterns for the No Letter group. The top panel presents detailed statistics for the pre-treatment period, during which 52% contributed to Obama and the remaining to Republican candidates. On average, individuals contributed about $524 during the pre-treatment period. Republican contributed substantially larger amounts than their Democratic counterparts, which was expected given that the Republican candidates were taking part in a primary while President Obama was only in the general election. The bottom half of Table B.3 presents similar statistics for post-treatment contributions. During the post-treatment period, 48.9% of our subjects made at least one contribution. For those who made contributions during the post-treatment period, the average amount contributed was $587. There are significant differences in post-treatment contributions across parties: e.g., the probability of making a post-treatment contribution was 75.9% for supporters of Obama, but only 38.6% for Romney supporters and 11.9% for supporters of other Republican candidates. Because of these differences in baseline rates, it is not straightforward to compare the magnitude of the effects across party lines.

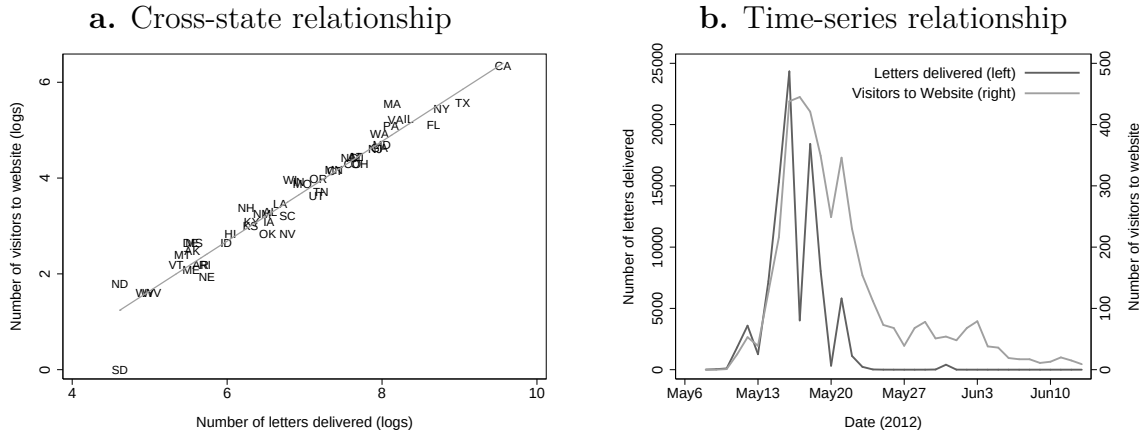
B.3 Further Details about the Table of Contributors Shown in the List Letter

As described in the body of the paper and illustrated by a facsimile in the previous section, the List letter type included a list of 9 neighbors along with the amount contributed and party contributed

to. Section 2.2.2 describes the methodology used to create random non-deceptive variation in this list of itemized contribution records. The set of constants ϵ_i's were drawn from the distribution $U(0, 10)$, plus an additional constant that takes an arbitrarily-large value for individuals living in the same household than the recipient (if any). Last, we set parameters c_p and c_a equal to one over the standard deviation of the share of democrats and the amount contributed, respectively.

Table B.4 presents summary statistics about the contribution records shown in that table, such as the average amount contributed by own- and opposite-party neighbors, as well as the number of neighbors contributing to the party of the recipient. We computed those statistics using the lists of contributors that would have been produced if we had set all of the weighting parameters to zero. That is, the variation in the contribution behavior corresponds to the baseline lists. As a result, Table B.4 illustrates the "natural" variation in contribution records and does not include the "induced" variation due to the randomization of the weighting parameters.

Figure B.1: Relationship Between the Mailing Delivery Indicator and the Number of Visits to the Project's Website



Notes: Date of delivery provided by USPS. Number of visitors to the website includes unique visitors that reached the website directly (approximately 83% of the visits) or indirectly through a search engine (in virtually all cases after searching for "www.campaign-information.info" or "campaign-information.info").

Table B.1: Comparison of Individual Characteristics for Individuals in the Subject Pool, for All Contributors in the 2012 Election Cycle and for the General U.S. Population

	Subject Pool	All 2012 Contributors	U.S. Averages
Percent Democrat	52.59	64.55	51.40
	(0.11)	(0.05)	
Mean Amount Contributed ($)	811.69	559.39	
	(2.01)	(0.82)	
Percent Male	59.37	55.15	49.14
	(0.11)	(0.05)	
Percent White	79.14	78.77	62.99
	(0.05)	(0.02)	
Percent Black	11.95	12.01	12.07
	(0.03)	(0.01)	
Population Density in ZIP-5	6.17	6.36	3.9
	(0.04)	(0.02)	
Mean Household Income ($1,000s), ZIP-5 average	105.01	98.09	55.24
	(0.26)	(0.11)	
Observations	191,832	1,070,098	

Notes: Average individual characteristics (standard deviations in parenthesis). The first column corresponds to individuals who made contributions to presidential campaigns from April 1, 2011 to April 1, 2012 and were selected for the field experiment according the criteria described in Section 3. The second column corresponds to all individuals who made contributions to presidential campaigns during the 2012 election cycle (the subject pool in the first column is a subset of this group). The third column corresponds to country-averages using the ZIP code level 2010 U.S. Census data. Data on contributions from the FEC public records, which includes individuals contributing over $200 to a campaign committee. The FEC database does not report information about the gender or the ethnicity of individual contributors. However, we constructed proxies for these variables based on information provided by the U.S. Bureau of the Census, which reports the joint distribution of first names and gender, and the joint distribution of last names and ethnicities. Population density and mean income come from 2010 U.S. Census data. The U.S. average share of democrats corresponds to the share of Democrat votes in the 2008 presidential election.

Table B.2: Balance of Observable Individual Characteristics across Treatment Groups

	(1) No-Letter	(2) Placebo	(3) Website Self	(4) Website Neighbors	(5) List Once	(6) List Update	(7) Difference Test P-value
Percent Democratic	52.59	52.67	53.33	52.53	52.46	52.00	0.24
	(0.16)	(0.37)	(0.37)	(0.37)	(0.37)	(0.37)	
Pre-treatment Contribution ($)	524.24	520.15	516.55	517.57	528.91	526.30	0.29
	(2.00)	(4.51)	(4.56)	(4.70)	(4.74)	(4.76)	
Percent Male	59.26	59.59	58.66	59.68	59.54	59.91	0.19
	(0.16)	(0.37)	(0.37)	(0.37)	(0.37)	(0.37)	
Percent White	79.12	78.99	79.17	79.20	79.20	79.32	0.82
	(0.07)	(0.17)	(0.17)	(0.17)	(0.17)	(0.17)	
Percent Black	11.91	12.11	11.86	11.98	12.10	11.90	0.40
	(0.05)	(0.11)	(0.11)	(0.11)	(0.11)	(0.11)	
Percent Hispanic	4.12	4.16	4.03	4.06	3.84	3.99	0.23
	(0.05)	(0.11)	(0.11)	(0.11)	(0.10)	(0.11)	
Observations	99,834	18,430	18,314	18,459	18,396	18,399	

Notes: Averages for different pre-treatment individual-level characteristics for treatment groups. Standard deviations in parenthesis. The last column reports the p-value of a test in which the null hypothesis is that the mean is equal for all the treatment groups. Data on amount contributed and recipient party from FEC public records (see Table B.3 for some descriptive statistics of this data). Ethnicity and sex were imputed according to first and last name frequencies reported by the U.S. Census Bureau.

Table B.3: Pre-Treatment and Post-Treatment Contribution Patterns, No-Letter Group

	All	Pre-treatment contribution to		
		Obama (DEM)	Romney (REP)	Others (REP)
Pre-Treatment Period				
Amount Contributed ($)	524.24	372.79	649.45	706.58
	(633.30)	(367.40)	(673.24)	(842.96)
Post-Treatment Period				
Percent Contributed at Least Once	48.91	75.93	38.61	11.86
	(49.99)	(42.75)	(48.69)	(32.33)
Amount Contributed ($), if Positive	586.97	568.16	600.29	753.04
	(681.77)	(644.95)	(716.40)	(921.02)
Observations	99,834	52,316	12,971	34,547

Notes: Average contribution behavior with standard deviations in parenthesis. The pre-treatment period spans from April 1, 2011 to the date the letter was received, and the post-treatment period spans from the date the letter was received to December 31, 2012. The other Republican candidates are: Bachman, Cain, Gingrich, Huntsman, Pawlenty, Perry and Santorum. Data from FEC public records.

Table B.4: Informational Effects: Summary Statistics

	Mean (Sd)
$\bar{c}_{\text{own}}$	6.33 (4.88)
$\bar{c}_{\text{opp}}$	5.92 (5.42)
N_{own}	5.31 (2.22)
$\sum c_{\text{own}} - \sum c_{\text{opp}}$	11.07 (48.97)
$\left\lvert \sum c_{\text{own}} - \sum c_{\text{opp}} \right\rvert$	36.70 (34.25)

Notes: $N = 36,795$. Summary statistics corresponding to the sample of individuals assigned to the List treatment. The descriptive statistics for these variables correspond to the counterfactual baseline letter (i.e., when all the weighting parameters used to generate the list of 9 neighbors are set to zero). All these variables except N_{own} are expressed in hundreds of dollars. $\bar{c}_{own}$ (conversely, $\bar{c}_{opp}$) corresponds to the average contribution of all the individuals in the list who contributed to the recipient's own (opposite) party. N_{own} is the number of individuals in the list who contributed to the recipient's party. $\sum c_{own} - \sum c_{opp}$ is the difference between total contributions to the own- and opposite-party. $|\sum c_{own} - \sum c_{opp}|$ is the absolute value of $\sum c_{own} - \sum c_{opp}$.

C Additional Results and Robustness Checks

C.1 General Robustness and Specification Checks

C.1.1 Robustness of Results to Alternative Regression Models and Standard Errors

This section presents a series of robustness and specification checks of the baseline results for the conformity and the comparison channels.

Table C.1 presents the main results for the conformity and comparison channels with the baseline specification (heteroskedasticity-robust standard errors with no clustering, column 1), and with clustering at different levels. The top panel reports the results for the conformity channel, and the bottom half reports the results for the comparison channel. Column (2) presents alternative results with clustering at the ZIP-5/party level (i.e., grouped by individuals from the same party and area of residence defined by the ZIP-5). In column (3), the standard errors are clustered at the ZIP-5 level, in column (4) at the ZIP-3 level, and in column (5) at the state level. The standard errors are very similar under all the alternatives, and the small differences do not imply changes in significance with respect to standard levels. The only noticeable difference is the clustering at the state level (column), which reduces standard errors and results in even more statistically significant coefficients.

Table C.2, in turn, presents the results for regression specification checks. The top panel reports the results for the conformity channel, and the bottom half reports the results for the comparison channel. Column (1) reports the baseline results (discussed in the body of the paper) which were computed by means of an interval regression model. Column (2) reports the results from the Tobit model. The point estimates and standard errors are virtually identical, for both the comparison and the conformity channels. In turn, we report in column (3) the estimates based on a Poisson regression model (the results – not reported – are virtually identical with a Negative Binomial model). In terms of statistical significance, the results are robust between columns (1) and (3). Quantitatively, we can compare the magnitude of these effects with the interval regression model from column (1). We start with the coefficient on Higher Visibility for the conformity channel. Those who made contributions during the post-treatment period gave on average \$587. The coefficient of -\$89.069 from column (1) corresponds to a –15.17% effect on the mean baseline contribution. Column (3) presents the simple coefficients from a Poisson regression. These coefficients can be interpreted as semi-elasticities: i.e., the coefficient of -0.177 implies that, in a place where everyone supports the opposite-party, the higher visibility intervention would decrease the expected amount contributed by 19.4% (i.e., $exp(0.177) - 1$). This is close to (and statistically indistinguishable from) the 15.17% implied by column (1). The results are similar for the coefficient on the interaction between *Higher Visibility* and *Share Own-Party*.

The bottom panel in Table C.2 presents the corresponding regression specification checks for the comparison channel. The Poisson estimates imply that a \$100 increase in $\bar{c}_{own}$ raises contributions by 0.6%. The interval regression estimates, in turn, imply that a \$100 increase in $\bar{c}_{own}$ raises contributions by 0.55% (i.e., 3.2/587) of the mean baseline rate. These two effects are also very close and statistically indistinguishable. The pattern of comparison of results is similar for the coefficient on N_{own}. In summary, if anything, the effect sizes implied by the Poisson model are slightly larger than the effect sizes implied by the interval regression model.

The last two columns of Table C.2 report results for the falsification test using the pre-treatment contributions. Column (5) corresponds to the specification reported in the body of the paper, which uses an OLS model with the amount contributed pre-treatment as the dependent variable. There is no point in showing the results for the interval regression model or the Tobit model because they would result in exactly the same estimates, since by construction all subjects in the sample had contributed non-zero amounts in the pre-treatment period. We can also verify if the falsification test is robust if we compute proportional effects instead. Column (6) presents the results when we use the logarithm of pre-treatment contributions as the dependent variable. The results are qualitatively similar: all the coefficients are close to zero, precisely estimated and not statistically significant. The results are also quantitatively similar. For example, the coefficient on *Higher Visibility* for the conformity effects in the linear model implies an effect of -0.5% (i.e., -3.316/587), while the logarithmic model implies an effect of 1.1%. Both coefficients are not statistically significant, and also statistically indistinguishable from each other. Most importantly, both estimates are very small compared to the magnitude of the effects on the post-treatment contributions (in the order of 15%-19%).

Table C.4 presents further robustness checks. The top panel reports the results for the conformity channel, and the bottom half reports the results for the comparison channel. Column (1) reports the baseline results (discussed in the body of the paper). Column (2) uses the same specification except that it excludes the control variables. Column (3) uses the same specification from column (1) with the only difference that it includes the observations for individuals with undeliverable mailings. As expected, the results are very similar across the three specifications.

C.1.2 Effects on Non-Presidential Contributions

We can also assess whether our treatments affected contributions to other federal campaigns besides the presidential campaign (e.g., Senatorial races). For the conformity channel, we would expect that a Democrat contributor in a highly Republican area should feel pressure to contribute less to all Democratic candidates, not only Obama. The top panel of column (4) in Table C.2 reports the results of a regression with the same specification as in column (1), but with the post-treatment amount contributed to non-presidential committees as the dependent variable (this information

was also obtained from the FEC public records).[ii] This additional outcome has the disadvantage that it has less variation in our subject pool: e.g., only 14.4% of our subjects contributed to non-presidential campaigns during the post-treatment period, whereas 49% contributed to presidential campaigns.[iii] The estimates from columns (1) and (4) (top panel) are very similar, suggesting that the effect of the higher visibility treatment on non-presidential contributions may well be similar to its effect on presidential contributions. However, due to the anticipated lack of precision, the estimates from column (4) are are not statistically significant at conventional levels.[iv]

The bottom panel of column (4) in Table C.2 reports the results of a regression with the same specification as in column (1) for the main specification of the comparison effects. The coefficients on $\bar{c}_{own}$, $\bar{c}_{opp}$ and N_{own} have the same sign in the bottom panel of column (4) and in column (1), suggesting that the information about presidential contributions may have affected non-presidential contributions in the same direction than it affected presidential contributions. However, due to lack of variation in the dependent variable, the coefficients from column (4) are much less precisely estimated and as a result they are not statistically significant at conventional levels.

C.1.3 Robustness to the Inclusion of Contributors to the Rand Paul Campaign

As discussed in the main body of the paper, subjects contributing to Rand Paul in the pre-treatment period made virtually no contributions to the Republican nominee that emerged from the primaries (i.e., Romney) in the post-treatment period. In the No-Letter group, only 1.56 percent of Rand Paul supporters contributed to Romney. This average contribution rate is an order of magnitude lower than the rest of the subjects (56 percent). The linear regressions used for the analysis of our main results do not allow the magnitude of the effects to vary with the baseline contribution rate, and this can be problematic for dealing with these very large differences in baseline contribution rates. Additionally, this pattern suggests that Rand Paul supporters may not identify themselves with the Republican party, which is a basic assumption for our conformity and comparison channel hypotheses. For all these reasons, we removed Rand Paul supporters from the database used for our main analysis and present our main results excluding this group.

Table C.3 presents the results of our main baseline results including and excluding these observations. The main results for both the conformity and the comparison channels remain unchanged

[ii] For the sake of simplicity, this measure excludes contributions to non-presidential committees or candidates from a party different from the one to which the presidential contributions were made (such cross-party variations in contributions are a very small minority: less than 1% of non-presidential contributions are made to another party).

[iii] Due to FEC reporting requirements, the proportions here, as in the rest of the paper, represent only individuals contributing more than the $200 threshold.

[iv] Another minor concern regarding post-treatment contributions is that a small number of subjects made presidential campaign contributions to both parties. In our results, we imputed them as contributors to the party to which they donated in the pre-treatment period. While we could present robustness checks regarding this assumption, these cases represent only 0.04% of our subjects: their number is so small that the results remain virtually identical if we drop these observations or if we make alternative assumptions.

under these alternatives samples: all the coefficients have the same sign and the same level of statistical significance with and without Rand Paul supporters (columns (1) and (2), and (3) and (4)). Of course, the post-treatment contribution behavior of Rand Paul supporters implies an attenuation in the effects, but while some of the coefficients are lower in absolute value when we include this group in the regressions, the differences are relatively small and the coefficients are statistically indistinguishable between the two sets of regressions. Finally, as expected from the random assignment process, the balance check for pre-treatment contributions is also qualitatively not affected by the inclusion or exclusion of Rand Paul supporters (columns (5) and (6)).

C.2 Conformity Channel: Heterogeneity with Respect to Other ZIP-3 Characteristics

In the robustness checks of the results for the conformity channel in Section 4.1, we tested whether the effect of higher visibility varied with the share of individuals of the recipient's own race, and with the share of low income households in the same ZIP-3 (Table 2). Here we present additional robustness checks to gauge whether the effects of higher visibility vary as a function of other important characteristics of the recipient's ZIP-3 population composition: race, education, age and marital status, as well as the area's population density and the average tax paid according to the 1040 IRS forms.

Results are reported in Table C.5, where we interact *Higher Visibility* with these other characteristics of the recipient's area of residence. The first column repeats the baseline specification with *Share Own-Party* as in column (1) in Table 2. In column (2) we present the coefficient of the interaction of *Higher Visibility* with the share of African Americans in the recipient's area of residence. This coefficient is very small (1.644) and not statistically significant at standard levels (SE of 77.164). We obtain the same qualitative result in columns (3) through (6) from Table C.5, in which we interact *Higher Visibility* with the share of college graduates, the share of individuals under 25 years old, the share of married individuals, the area's population density and the average tax rate. In none of these specifications the coefficient of *Higher Visibility* nor that of its interaction with these area's characteristics are statistically significant at standard levels, and they are also economically small (close to zero).

Since all these variables have different distributions, the best way to compare the magnitude of the coefficients is by studying the effect of a one standard deviation increase in each ZIP-3 characteristic. The coefficients imply that a one standard deviation would increase the effect of *Higher Visibility* by \$24.39** (SE 10.62) for *Share Own-Party*, \$0.19 (SE 9.49) for *Share Black*, \$3.02 (SE 10.10) for *Share College*, \$1.09 (SE 11.23) for *Share <25yo*, \$0.25 (SE 9.99) for *Share Married*, -\$5.79 (SE 9.50) for *Population Density* and -\$3.95 (SE 12.37) for *Mean Tax Rate*. The

pairwise differences of these coefficients with that of the interaction of *Higher Visibility* and *Share Own-Party* are either statistically significant or marginally non-significant. In other words, there is no evidence of heterogeneity in the impact of the higher visibility treatment with respect to other ZIP-3 characteristics. Finally, our main results remain virtually unchanged if we estimate the specification in column (1) (i.e., with the interaction of *Higher Visibility* and *Share Own-Party*) and include additional controls for the area characteristics and their interactions with *Higher Visibility* (results not reported).

C.3 Comparison Channel: Additional Results

C.3.1 Results with Alternative Statistics

In Section 4.2 we reported that a higher number of own-party contributors (N_{own}) discourages contributions. In this section, we start by exploring whether the effect from N_{own} may be due to contributors updating their beliefs about the probability of making a pivotal contribution.

Some individuals may contribute because of the perception that, with some probability, their marginal contribution will change the election outcome (Ansolabehere et al., 2003). This is similar to the probability of being a pivotal voter in a model of voter turnout (e.g., Dhillon and Peralya, 2002).[v] One common argument against this theory, similar to the argument about a marginal vote, is that the average individual contribution of a few hundred dollars is infinitesimal when compared to the several hundreds of millions of dollars raised by each candidate, so that the probability of making a pivotal contribution is extremely small. However, individuals may still systematically over-estimate this probability. One prediction from this line of reasoning is that individuals should be more likely to contribute when an election is close. Since the distribution of contributions between the two parties may be a signal of how close an election will be, a prediction from this theory is that contributors should care about the contributions of others. For example, if we assume that a close campaign signals a close election, then a contributor should be more eager to make a contribution when she observes that the "contribution race" is more even. Note that, however, the List letter included contribution records from the individual's area of residence, so that this could affect the perception of making a pivotal contribution only to the extent that individuals extrapolate how the campaign is going from the local to the national level.

The regression results are presented in Table C.6. Column (1) reproduces the results from the baseline specification in Section 4.2. Column (2) replaces N_{own} by the difference in total contributions between both parties, $\sum c_{own} - \sum c_{opp}$.[vi] As in column (1), the significant negative

[v] Dhillon, A. and Peralta, S. (2002), "Economic Theories of Voter Turnout," The Economic Journal, Vol. 112 (480), pp. F332-F352.

[vi] Note that, in this specification, increasing $\bar{c}_{own}$ while holding N_{own} constant has an effect through both $\bar{c}_{own}$ and $\sum c_{own}$.

coefficient suggests that recipients make less generous contributions when they are shown that their own party is doing better than the opposite party. To disentangle the pivotal contribution motive, the specification in column (3) includes as an additional independent variable the absolute value of the difference between the total amounts contributed to the recipient's own and opposite-party, $|\sum c_{own} - \sum c_{opp}|$, which is a measure of how uneven the campaign is. The coefficient on this variable has the expected sign: individuals are less motivated to make a contribution in more uneven races. However, this coefficient is smaller than the coefficient on $\sum c_{own} - \sum c_{opp}$, and it is not statistically significant.[vii] Furthermore, the coefficient on $\sum c_{own} - \sum c_{opp}$ remains similar to the corresponding coefficient from column (2). Finally, for completeness, column (4) in Table C.6 presents the falsification test with pre-treatment contributions as the dependent variable. As in all other results for this type of falsification test, none of the included variables have a significant effect on pre-treatment contributions.

C.3.2 Robustness Check: Controlling for Area Fixed Effects

One concern is whether the estimated comparison effects are due solely to the experimental assignment, or whether they may be contaminated by cross-sectional differences in the neighbors of the recipients.

In the body of the paper, we discuss a key falsification test: we estimate the same regression with pre-treatment contributions instead of the post-treatment contributions as the dependent variable (Table 3, column 6). In this subsection, we present an additional robustness check, consisting of estimating our baseline regression with ZIP-5 fixed effects. Intuitively, two individuals in the same ZIP-5 are more likely to have a very similar list of 30 neighboring contributors that could potentially end up in the letter we mailed. Our experimental assignment procedure provides these two individuals from the same ZIP-5 with different subsets of contributors from the same area. If the identification is due to the experimental assignment, then controlling for these fixed effect should not affect the results in a meaningful or substantial way. If, however, the identification was contaminated by non-experimental differences across space, then the effects should disappear or change substantially with the inclusion of the area fixed effects.

The results from this exercise are presented in Table C.7. One technical challenge is that, to the best of our knowledge, there is no fixed-effects estimator for an Interval Regression model. Instead, we use the Tobit model as a baseline, which provides very similar results to the Interval Regression model (as shown in Appendix C.1) and for which there is a fixed-effects estimator (Honoré, 1992).[viii] Column (1) presents the baseline results from the Tobit model without ZIP-5

[vii] It is also possible that the investment motive would have been more relevant if, instead of generating variation in how close the campaign was, we had created variation directly on how close the election was expected to be (such as using information from prediction markets).

[viii] Honoré, B.E. (1992), "Trimmed Lad and Least Squares Estimation of Truncated and Censored Regression

fixed effects, while column (2) presents results including the fixed effects. We cannot reject the null hypothesis that the coefficients in column (1) and (2) are equal. If anything, the coefficients are larger in absolute value with the addition of the fixed effects, contrary to what we would expect from non-experimental "contamination" of our results. However, we should not over-interpret this difference, because the semi-parametric estimator from Honoré (1992) results in less precisely estimated coefficients.

We can provide an additional robustness check using the Poisson model, for which there is also a fixed-effects estimator. Column (3) presents the baseline Poisson results without ZIP-5 fixed effects. As discussed in Appendix C.1, the sign, statistical significance and magnitude of the effects are similar between the Poisson and Tobit estimates. Column (4) presents results of a similar model including the ZIP-5 fixed effects. The coefficients are almost identical in columns (3) and (4), suggesting again that controlling for fixed-effects does not make a tangible difference in our results. In sum, the evidence from this robustness check confirms the hypothesis that the identification is coming from the experimental variation.

C.3.3 Results on the Effects of Disseminating Unbiased Information about Contribution Records

In this section, we study whether the dissemination of objective information about the contribution behavior of others may affect contributions, possibly due to systematic biases in the perceptions about the contribution behavior of others.

We measure the effect of providing contributors with unbiased information about contribution records of other individuals in their area (i.e., the table of contributions contained in the List letter). If individuals had a systematically biased perception of the contribution behavior of others, then the unbiased information should have a significant effect on their contributions. For instance, suppose that individuals, on average, under-estimate the average amount contributed by other supporters of their same party.[ix] Since the individual's contribution is increasing in this belief (see Section 4.2), then the distribution of unbiased information should have a positive effect on the contributions of the recipients.

However, the comparison of contributions between the List and No-Letter subjects may be contaminated by other pieces of information contained in the List letter that are unrelated to the information about contribution behavior of others. For instance, simply receiving a letter about campaign contributions may remind individuals about their commitment to contribute, and this could have a positive effect on future contributions independently of the information provided in the

Models with Fixed Effects," Econometrica, Vol. 60 (3), pp. 533–565.

[ix] There are several reasons why individuals may fail to form an accurate perception of the distribution of contributions. Tversky, A and Kahneman, D. (1973), "Availability: A heuristic for judging frequency and probability," Cognitive Psychology, Vol. 5 (1), pp. 207–233.

body of the letter. Alternatively, receiving a letter from a research team may have an effect of its own, for example by making the individual think that campaign contributions are more important than previously thought. To test this hypothesis, we use the Placebo letter, which has the same format than the List letter but, instead of providing a table about the contributions of neighbors, it displayed standard regulatory information about contribution limits. We did not expect this information to have an effect on contributions, because contribution limits were not binding for virtually all of the individuals in the No-Letter group. The Placebo letter could still have some effects if, for example, it works as a reminder that the individual must make a contribution, because of the feeling that the recipient is part of an academic study, etc. By comparing the effects of the List and Placebo letters, we can provide some suggestive evidence about the effect of the List letter that can be attributed exclusively to the table with information about contributions of others, rather than to these other confounding factors.

There is, however, a second limitation with this comparison. We explicitly took measures to ensure that recipients of the List letter would not feel more observed by their neighbors: e.g., semi-anonymizing the records, not including information about the search tool of the FEC website, choosing neighbors from a broader geographical area. Despite our efforts, however, it is possible that part of the effect of the List letter, compared to the Placebo letter, came from the recipient feeling more observed by neighbors. In principle, this should not be a problem because, as seen in the results about conformity effects, the average effect of increased visibility is practically zero. However, since we did not know this fact before running the experiment, we randomized another aspect of the List-letter to test whether it significantly increased the perceived visibility of the recipient's contribution. We randomly assigned subjects to one of two variations of the List treatment: List-Once (illustrated in Appendix A.3) and List-Update (Appendix A.4). The only difference between these two variations was that in the List-Update letter we stated that an updated list with contributions by neighbors could be sent at the end of the election cycle, whereas the List-Once letters specified that a letter of this type would not be sent again in the future. If individuals felt that their names could be recognized by neighbors they knew, they should feel more monitored in terms of their future contributions from the List-Update letter, because the neighbors would get an update about the subsequent contributions.

Table C.8 presents a series of comparisons between the post-treatment amounts contributed in these different treatment groups. Column (1) shows that, compared to the No-Letter group, sending an individual a List letter increases the amount contributed post-treatment by about \$19.45. This effect is statistically significant at the 1% level, and also economically significant. Recall that, among those who contributed at least once during the post-treatment period, the mean amount contributed was about \$587. This means that the effect of the List letter accounted for 3.3% of this baseline contribution. Column (4) shows that the effect of the List letter on

the probability of making a post-treatment contribution was about 0.478 percentage points (p-value<0.10). Once again, the effect on the extensive margin was somewhat lower than on the amount contributed: this effect of the List letter amounted to just about 0.85% of the baseline rate of 55.72% (i.e. the mean probability of post-treatment contribution in the No-Letter group). As a falsification test, column (5) estimates the "effect" of the List letter on pre-treatment contributions. As expected, the coefficient is very close to zero and not statistically significant.

Columns (2) and (3) explore whether the effects of the List letter were indeed due to the information about the contribution patterns of others. Instead of comparing the List and No-Letter groups, column (2) offers a comparison between the List and Placebo groups. The coefficient is similar in magnitude ($15.82) to the corresponding coefficient from column (1) ($19.45), and it is also statistically significant. Furthermore, we cannot reject the null hypothesis that these two coefficients are equal. Another way of looking at this evidence is that, compared to the No-Letter group, the Placebo did not affect the contribution behavior of recipients. Last, column (3) compares post-treatment contributions between the List-Update and List-Once groups. The difference between these two is very close to zero and not statistically significant. This result suggests that the effect of the List letter cannot be attributed to increased visibility of the recipient's contribution.

In sum, the evidence from this appendix suggests that the information about contribution records from the List letter may have corrected some systematic biases in the recipients' perception about the contribution behavior of others. More precisely, the positive effect on contributions may imply that individuals systematically under-estimate the mean amount contributed by own-party neighbors, that they over-estimate the number of own-party contributors relative to the opposite-party contributors, or a combination of both.[x]

[x] In unreported results we also find that the effect of the List letter was more positive in areas with higher average contributions by own-party neighbors and in areas where the total contributions to the own-party are surpassed by the total contributions to the opposite-party.

Table C.1: Robustness to Different Alternatives of Standard Error Clustering

	Dep Var.: Amount Contributed Post-Treatment				
	(1)	(2)	(3)	(4)	(5)
Conformity Channel:					
Higher Visibility	-89.069**	-89.069**	-89.069**	-89.069**	-89.069***
(Website-Neighbors - Website-Self)	(37.222)	(38.295)	(38.147)	(40.077)	(30.262)
Interaction with:					
Share Own-Party in ZIP-3	143.499**	143.499**	143.499**	143.499**	143.499***
	(60.303)	(62.493)	(62.099)	(65.406)	(50.163)
Comparison Channel:					
$\bar{c}_{\text{own}}$	3.200**	3.200**	3.200**	3.200**	3.200**
	(1.473)	(1.459)	(1.446)	(1.444)	(1.387)
$\bar{c}_{\text{opp}}$	-0.399	-0.399	-0.399	-0.399	-0.399
	(0.956)	(0.955)	(0.964)	(1.003)	(1.171)
N_{own}	-5.440*	-5.440*	-5.440*	-5.440*	-5.440**
	(2.862)	(2.899)	(2.886)	(2.830)	(2.706)
SE Clustered By:	None (Robust)	ZIP-5/Party	ZIP-5	ZIP-3	State

Notes: $N = 32,070$ (top panel, conformity channel) and $N = 31,996$ (bottom panel, comparison channel). * significant at the 10% level, ** at the 5% level, *** at the 1% level. Column (1) presents results with heteroskedasticity-robust standard errors in parenthesis. Column (2) presents alternative results with clustering at the ZIP-5/party level (i.e., grouped by individuals from the same party and area of residence defined by the ZIP-5). In column (3), the standard errors are grouped by ZIP-5 areas, in column (4) by ZIP-3, and in column (5), by state. Control variables and details of regression specifications as in Tables 2 and 3 in the body of the paper.

Table C.2: Robustness Checks: Alternative Regression Models and Specifications

	Post-Treatment Contributions				Pre-Treatment	
	(1)	(2)	(3)	(4)	(5)	(6)
	$	$	$	$	$	log($)
Conformity Channel:						
Higher Visibility	-89.069**	-89.025**	-0.177**	-52.887	-3.316	0.011
(Website-Neighbors - Website-Self)	(37.222)	(37.225)	(0.074)	(263.288)	(21.766)	(0.036)
Interaction with:						
Share Own-Party in ZIP-3	143.499**	143.453**	0.279**	93.527	3.365	-0.019
	(60.303)	(60.308)	(0.117)	(430.402)	(34.163)	(0.060)
Comparison Channel:						
$\bar{c}_{\text{own}}$	3.200**	3.197**	0.006**	9.499	0.896	0.000
	(1.473)	(1.473)	(0.003)	(11.916)	(0.910)	(0.001)
$\bar{c}_{\text{opp}}$	-0.399	-0.401	-0.000	-5.361	0.061	0.000
	(0.956)	(0.956)	(0.002)	(6.957)	(0.580)	(0.001)
N_{own}	-5.440*	-5.447*	-0.011**	-19.645	1.586	0.003
	(2.862)	(2.862)	(0.005)	(19.852)	(1.750)	(0.003)
Regression Model	Interval	Tobit	Poisson	Interval	OLS	OLS
Contribution Type	Pres.	Pres.	Pres.	Non-Pres.	Pres.	Pres.

Notes: $N = 32,070$ (top panel, conformity channel) and $N = 31,996$ (bottom panel, comparison channel). * significant at the 10% level, ** at the 5% level, *** at the 1% level. Heteroskedasticity-robust standard errors in parenthesis. Control variables and details of regression specifications as in Tables 2 and 3 in the body of the paper. The contribution type *Pres.* corresponds to presidential committees (i.e., Obama and Romney), while *Non-Pres.* corresponds to non-presidential committees.

Table C.3: Robustness to the Inclusion of Contributors to the Rand Paul Campaign

	Post-Treatment Contributions				Pre-Treatment	
	(1)	(2)	(3)	(4)	(5)	(6)
	$	$	P($)>0	P($)>0	$	$
Conformity Channel:						
Higher Visibility	-89.07**	-77.17**	-3.53**	-2.69**	-3.32	-11.02
(Website-Neighbors - Website-Self)	(37.22)	(36.72)	(1.61)	(1.33)	(21.77)	(19.90)
Interaction with:						
Share Own-Party in ZIP-3	143.50**	128.82**	6.67**	5.40**	3.37	16.00
	(60.30)	(59.69)	(2.71)	(2.34)	(34.16)	(31.97)
Observations	32,070	36,773	32,070	36,773	32,070	36,773
Comparison Channel:						
$\bar{c}_{\text{own}}$	3.20**	2.88**	0.05	0.04	0.90	0.74
	(1.47)	(1.46)	(0.06)	(0.05)	(0.91)	(0.82)
$\bar{c}_{\text{opp}}$	-0.40	-0.61	-0.05	-0.05	0.06	0.11
	(0.96)	(0.95)	(0.04)	(0.04)	(0.58)	(0.57)
N_{own}	-5.44*	-6.15**	-0.16	-0.17	1.59	2.04
	(2.86)	(2.85)	(0.13)	(0.11)	(1.75)	(1.64)
Observations	31,996	36,795	31,996	36,795	31,996	36,795
Incl. Rand Paul	No	Yes	No	Yes	No	Yes
Regression Model	Interval	Interval	OLS	OLS	OLS	OLS

Notes: * significant at the 10% level, ** at the 5% level, *** at the 1% level. Heteroskedasticity-robust standard errors in parenthesis. Control variables and details of regression specifications as in Tables 2 and 3 in the body of the paper.

Table C.4: Robustness to the Exclusion of Control Variables and the Inclusion of Undeliverable Mail

	Post-Treatment Contributions		
	(1)	(2)	(3)
	$	$	$
Conformity Channel:			
Higher Visibility (Website-Neighbors - Website-Self)	-89.07**	-98.20**	-87.79**
	(37.22)	(38.64)	(37.06)
Interaction with:			
Share Own-Party in ZIP-3	143.50**	146.32**	141.92**
	(60.30)	(62.70)	(60.07)
Comparison Channel:			
$\bar{c}_{\text{own}}$	3.20**	3.06**	3.09**
	(1.47)	(1.52)	(1.47)
$\bar{c}_{\text{opp}}$	-0.40	-0.29	-0.53
	(0.96)	(1.01)	(0.95)
N_{own}	-5.44*	-4.54	-5.55*
	(2.86)	(2.99)	(2.86)
Controls	Yes	No	Yes
Exclude Undeliverable	Yes	Yes	No

Notes: * significant at the 10% level, ** at the 5% level, *** at the 1% level. Heteroskedasticity-robust standard errors in parenthesis. Control variables and details of regression specifications as in Tables 2 and 3 in the body of the paper. Column (2) excludes the set of control variables. Column (3) includes observations for individuals with undeliverable mailings (366 observations in the estimation of the conformity channel and 352 observations for the comparison channel).

Table C.5: Heterogeneity of Higher Visibility with Other ZIP-3 Characteristics

	Dep Var.: Amount Contributed Post-Treatment						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Higher Visibility	-89.069**	-5.150	-22.344	-15.224	-6.428	-2.111	2.770
(Website-Neighbors - Website-Self)	(37.222)	(14.272)	(57.972)	(106.017)	(53.681)	(11.257)	(23.734)
Interaction with:							
ZIP-3 Characteristic	143.499**	1.644	30.240	21.717	2.864	-41.328	-30.351
	(60.303)	(77.164)	(98.965)	(222.212)	(111.666)	(71.426)	(88.715)
ZIP-3 Characteristic	Share Own-Party	Share Black	Share College	Share <25yo	Share Married	Pop. Density	Mean Tax Rate
Mean ZIP-3 Characteristic	0.57	0.13	0.58	0.48	0.48	0.07	0.26
Std. ZIP-3 Characteristic	0.17	0.12	0.10	0.05	0.09	0.14	0.13
Observations	32,070	32,070	32,070	32,070	32,070	32,070	32,070

Notes: * significant at the 10% level, ** at the 5% level, *** at the 1% level. Heteroskedasticity-robust standard errors in parenthesis. Observations from subjects assigned to Website letters. The dependent variable is the amount contributed during the post-treatment period. *Higher Visibility* is a dummy on whether the subject received a Website-Neighbors rather than a Website-Self letter. *Share Own-Party* stands for the share of own-party contributors to presidential campaigns in the ZIP-3 during the three previous presidential election cycles. The source for the following ZIP-3 characteristics is the American Community Survey 2012: *Share Black* refers to the share of African Americans in the recipient's ZIP-3, *Share College* is the share of college graduates in the same area, *Share <25yo* is the share of the population below 25 years of age in the area, *Share Married* is the share of married individuals in the area*, and Pop. Density* represents the population density. The *Mean Tax Rate* is the average tax rate paid in 1040 IRS forms (source: IRS Statistics of Income 2012). All the regressions include as controls the levels of all variables that are interacted with *Higher Visibility*, the time it took for delivery of the mailpiece, and a set of variables with pre-treatment contributions to each candidate. The contribution type *Pres.* corresponds to presidential committees (i.e., Obama and Romney), while *Non-Pres.* corresponds to non-presidential committees. *Mean Outcome* corresponds to the average of the outcome variable over the entire sample. Data on contributions from the FEC public records (see Table B.3 for descriptive statistics)..

Table C.6: Comparison Channel: Additional Specifications and Robustness Checks

	Post-Treatment Contributions						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	$	$	$	$	$	$	$
$\bar{c}_{\text{own}}$	2.877**	4.194***	4.679***	4.351***	1.452	2.223*	9.201
	(1.450)	(1.579)	(1.686)	(1.588)	(1.208)	(1.147)	(11.370)
$\bar{c}_{\text{opp}}$	-0.609	-1.426	-1.364	-1.529	-0.092	-0.547	-6.686
	(0.951)	(1.082)	(1.084)	(1.087)	(0.721)	(0.779)	(6.993)
N_{own}	-6.147**				-6.748***	-0.949	-25.199
	(2.883)				(2.304)	(2.271)	(19.494)
$\sum c_{\text{own}} - \sum c_{\text{opp}}$		-0.415**	-0.432**	-0.438**			
		(0.173)	(0.176)	(0.174)			
$\lvert\sum c_{\text{own}} - \sum c_{\text{opp}}\rvert$			-0.155				
			(0.179)				
$\lvert\sum c_{\text{own}} - \sum c_{\text{opp}}\rvert^{\star}$				-0.323			
				(0.201)			
Regression Method	Interval	Interval	Interval	Interval	Interval	Interval	Interval
Sub-Period					<Sep-1	≤ Sep-1	
Contribution Type	Pres.	Pres.	Pres.	Pres.	Pres.	Pres.	Non-Pres.
Mean Outcome	$295	$295	$295	$295	$147	$149	$292

Notes: $N = 31,996$. * significant at the 10% level, ** at the 5% level, *** at the 1% level. Heteroskedasticity-robust standard errors in parenthesis. Observations from subjects assigned to the List letter. All the independent variables except N_{own} are expressed in hundreds of dollars (i.e., the estimates correspond to effects of $100 changes in the independent variables). $\bar{c}_{own}$ (conversely, $\bar{c}_{opp}$) corresponds to the average contribution of all the individuals in the list who contributed to the recipient's own (opposite) party. N_{own} is the number of individuals in the list who contributed to the recipient's party. $\sum c_{own} - \sum c_{opp}$ is the difference between total contributions to the own- and opposite-party. $|\sum c_{own} - \sum c_{opp}|$ is the absolute value of $\sum c_{own} - \sum c_{opp}$. See Table B.4 for descriptive statistics for all these independent variables. These independent variables are included in the regression as the difference between the value computed with the list sent to the recipient and the corresponding value computed in the baseline list. *Mean Outcome* corresponds to the average of the outcome variable over the entire sample. All regressions include as control variables the time elapsed until mailing delivery and a set of variables with the pre-treatment contributions to each candidate. Data on contributions from the FEC public records (see Table B.3 for descriptive statistics).

Table C.7: Comparison Channel: Robustness to Controlling for Area Fixed Effects

	Dep Var.: Amt. Cont. Post-Treatment			
	(1)	(2)	(3)	(4)
$\bar{c}_{\text{own}}$	3.197**	4.836**	0.006**	0.006**
	(1.473)	(2.168)	(0.003)	(0.003)
$\bar{c}_{\text{opp}}$	-0.401	0.419	-0.000	0.000
	(0.956)	(1.462)	(0.002)	(0.002)
N_{own}	-5.447*	-7.497*	-0.011**	-0.010*
	(2.862)	(4.346)	(0.005)	(0.006)
Regression Model	Tobit	Tobit	Poisson	Poisson
ZIP-5 Fixed Effects?	No	Yes	No	Yes

Notes: $N = 31,996$. * significant at the 10% level, ** at the 5% level, *** at the 1% level. Heteroskedasticity-robust standard errors in parenthesis, except for column (2) which uses bootstrapped standard errors. Independent variables as in the previous Table.

Table C.8: Estimating the Effects of Disseminating Unbiased Information about Contribution Records

	Post-Treatment				Pre-Treatment
	(1)	(2)	(3)	(4)	(5)
	$	$	$	P($>0)	$
List vs. No-Letter	18.784***			0.416*	2.223
	(6.062)			(0.239)	(4.362)
List vs. List-Placebo		15.056*			
		(9.048)			
List-Update vs. List-Once			-0.428		
			(10.778)		
Observations	136,629	55,225	36,795	136,629	136,629
Regression	Interval	Interval	Interval	OLS	OLS

Notes: * significant at the 10% level, ** at the 5% level, *** at the 1% level. Heteroskedasticity-robust standard errors in parenthesis. The dependent variable in columns (1) through (3) is the amount contributed post-treatment. The dependent variable in column (4) takes the value 100 if the individual made at least one contribution post-treatment and 0 otherwise. The dependent variable in column (5) is the dollar amount contributed in the pre-treatment period. All regressions except column (5) include the usual set of individual-level control variables: time since mail delivery and a set of variables on pre-treatment contributions to each candidate.

D The Post-Election Mail-In Survey and the Scale Up of the Conformity Effects

D.1 More Details and Descriptive Statistics About the Survey

We collected complementary information from a subsample of our subjects by means of a mail-in survey sent after the election. The survey was sent by mail on December 6, 2012, one month after the 2012 presidential election day. The intended recipients, 44,380 in total, were a random sample of individuals from the No-Letter group (one third) and from the Website treatment group (two thirds).[xi] The survey instrument and its envelope are shown in the following pages. The envelope contained a letter and the survey on two sheets of paper, and a smaller prepaid business-reply envelope. The recipient was asked to fill out the survey and mail it back in the provided envelope by dropping it in a USPS mailbox. The individual could keep the separate letter, which contained details about the survey (e.g., confidentiality of the responses) as well as contact information for the research team. During the three months after we sent the surveys, we received 9,414 responses. This implies a response rate of 21.21%. It should be noted that there were significant differences in response rates for key sub-groups of the population. Most notably, the response rate for Democrats, at about 27%, was substantially higher than that of Republicans, at about 12%. The information from the survey discussed in the paper thus over-represents Democrat contributors. However, the mail-in survey response rate was 21.0% for subjects in the No-Letter group and 21.3% for recipients of the Website letter (difference not statistically significant – p-value of 0.357).

As an incentive for participation, the letter informed recipients that there were lottery prizes for individuals who responded and mailed back the survey before January 31, 2013. Half of the recipients were randomly assigned to be eligible for ten lottery prizes of $100 each, while the other half were eligible for ten lottery prizes of $200 each. The purpose of randomizing the stakes of the lottery was to provide some orthogonal variation in response rates that could be exploited to correct potential selection biases in the mail-in survey respondent pool. The response rate was only half a percentage point higher in the group eligible to the higher lottery prize, which implies an increase in the response rate of approximately 2.6%. This effect is relatively small and not statistically significant (p-value of 0.16). Such small effect of the lottery prize does not provide useful variation for the analysis. The fact that contributors did not react significantly to this economic incentive is probably due to the fact that the average recipient is relatively well-off and thus less sensitive to pecuniary incentives. Nevertheless, we must note that the odds of receiving a prize were low, which could have resulted in very small differences in the perceived expected value

[xi] As discussed in the paper, contributors to the Rand Paul primary made virtually no contributions in the post-treatment period to the Romney presidential campaign. We thus excluded this group from the sample for the mail-in survey.

of the prize.

Last, Figure D.1 shows the distribution of responses for two questions that were not included in the main body of the paper. Figure D.1.a shows the perceptions about the regulation of contribution limits, while Figure D.1.b shows the perceptions about contribution norms.

D.2 Further Details on the Scale-Up of the Conformity Channel Effects

D.2.1 The Identification Assumption and the Exclusion Restriction in the Estimation of the Conformity Channel

A straightforward way to think about the identification assumption for our conformity channel experiment is to consider the case of a binary treatment: "reading a letter a Website-Neighbors instead of a Website-Self letter" (R_i). We are interested in identifying the casual effect of this treatment. However, our experiment implies that we can only observe the effects of a related but different treatment, "being sent a Website-Neighbors instead of a Website-Self letter" (S_i). There are three possibilities regarding the relationship between R_i, S_i and contributions:

First, there may be an homogeneous effect of R_i on contributions, and an homogeneous effect of S_i on R_i. In this case, we do not need to make any assumptions other than the usual exclusion restriction for identifying the causal effect of Si. We can compute the TOT effect by means of a Wald estimate, where the numerator is the ITT estimated in our experimental analysis, and the denominator is the effect of S_i on R_i (i.e., the reading rate). Since we do not observe the latter effect, we construct a proxy based on the post-election mail-in survey data.

Second, there may be an homogeneous effect of R_i on contributions, but heterogeneous effects of S_i on R_i. In this case, the denominator of the Wald estimate should be the average effect of S_i on R_i for the entire sample of subjects (i.e., not only respondents to the mail-in survey). To use the results from the survey, we need to assume that the average propensity to read the letter in the sample of survey respondents reflects the average for the entire sample of experimental subjects. There are several plausible instances in which this assumption can be violated. The most likely scenario is that survey respondents have less limited attention (or a lower opportunity cost of time), so that the reading rate is higher among survey respondents than among non-survey respondents. However, this source of bias would lead to an over-estimation of the reading rate and, in turn, an under-estimation of the scaled-up effects, as discussed in the previous sections.

A third case would have heterogeneous effects of R_i on contributions, and heterogeneous effects of S_i on R_i. In this case, we need to make all of the assumptions from Imbens and Angrist (1994).[xii]

[xii] Imbens, G.W. and Angrist, J.D. (1994), "Identification and Estimation of Local Average Treatment Effects," Econometrica, Vol. 62 (2), pp. 467-475.

It should be noted that in our context we only have, by construction, compliers and never-takers, since we do not expect non-recipients to read the letter we sent to those selected for the treatment. As a result, the monotonicity assumption is satisfied automatically. We can also safely assume that sending someone a letter has no effect on the recipient if she does not actually read the letter. Under these conditions, the Wald estimate corresponds to the Local Average Treatment Effect of R_i – i.e., the effect on those who read the letter because we sent it to them, but would not have read it if we had not sent it. As in the previous case, to use the mail-in survey as a basis for the Wald estimate, we must rely on the assumption that the average propensity to read the letter in the sample of survey respondents is representative of the average reading rate for the entire sample of experimental subjects.[xiii]

D.2.2 Qualifications of the Scale Up of Effects

The discussion in the body of the paper indicated large scaled-up effects. We can, of course, qualify the results obtained with these simple estimates of the reading rate. A first concern is that the Website letter may have induced a lower willingness to participate in the mail-in survey, which could lead to an under-estimation of the reading rate. However, the mail-in survey response rate was 21.0% for subjects in the No-Letter group and 21.3% for recipients of the Website letter, and this 0.37 percentage points difference is not only very small but also not statistically significant (p-value of 0.357).

A second concern is that the mail-in survey sample may not be representative of the subject pool, and thus the estimated reading rate may not correspond to that of the overall sample. However, individuals who were more likely to read our letter were probably also more likely to respond to our survey. For instance, more pro-social subjects, or those who had more free time, may have been more likely to have read our Website letter and to have responded to our mail-in survey. This type of bias implies an over-estimation of the reading rate and, in turn, an under-estimation of the scaled-up effects.

A third concern is that our estimate of the reading rate is based on the assumption that all of the recipients of the Website letter who were unsure about the publicity of individual contributions went on to report that this information is public in the mail-in survey. Some of these recipients, however, might not have been induced to state that these records are public even after reading the letters. This implies that we could be under-estimating the actual reading rate. In any case, even with a reading rate half as large (and a scale-up factor half the size) as in our estimates, the conformity effects would still imply very large changes in contribution behavior.

[xiii] A different question is whether this LATE/TOT under- or over-estimates the ATE, which is recurrent discussion in papers relying on IV for the identification of causal effects. For example, "nicer" people may exhibit a higher propensity to read the letter, and they may also be more sensitive to social incentives. In this case, the TOT would be above the ATE.

Last, it should be noted that the mail-in survey's response rate, 21.2%, was relatively high, which suggests that the reading rate for the original treatment letters may have been even higher. In fact, this response rate was more than twice what we expected and used for our power calculations. However, the conditions of the survey mailing were very different than the conditions of the experimental mailing. First, the mail-in survey was sent in a closed envelope, whereas the experimental mailpiece consisted of a single sheet of paper that folded and sealed to make letter-sized mailpiece, and the latter design is more likely to be discarded unopened. Second, contributors received much more unsolicited physical and electronic mailing related to the election at the time we sent the experimental mailpieces, in the middle of the presidential campaigns. On the contrary, we sent the mail-in survey a month after the election, when subjects were not receiving any correspondence related to the campaign. This also implies that the mail-in survey envelope was substantially less likely to be discarded unread than the mailpieces corresponding to the experiment.

D.2.3 Alternative Estimate of the Scale Up of Effects

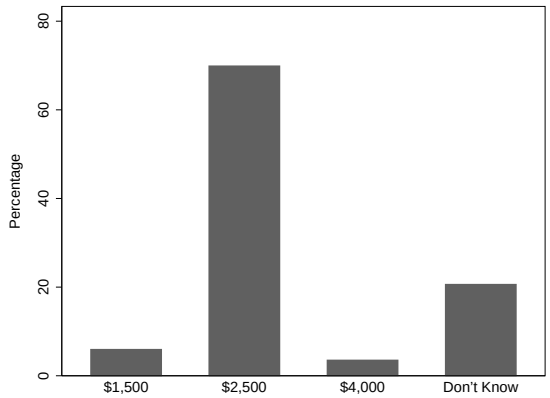
Section 4.1 presents results on the effect of the information in our treatment, with an estimate of the implicit reading rate of our letters of $r = 0.215$ (with a 90% confidence interval between 0.146 and 0.284), and a scale-up factor of 4.6. This estimate originates in the assumption that the Website letter reduced the proportion of those being unsure about their answers about the publicity of contributions in our post-election mail-in survey. Alternatively, we could assume that the Website letter decreased the proportion of those who did not know that contribution records were public: i.e., it shifted individuals from any response category to being somewhat sure or very sure about the publicity of contribution records. The share of respondents who did not select any of these two categories was 25.6% in the No-Letter group and 21.2% in the Website group. The difference between the two results in an implicit reading rate of $r = 0.171$ and a scale up factor of 5.8. This alternative estimate thus leads to an even higher scale-up factor. The estimate in the body of the paper and this alternative estimate are both within the range provided by the mass-mailing experts which we used for our power calculations.

One reason why these two estimates differ is that the Website letter did not affect some individuals who reported to be somewhat or very sure that campaign records were confidential. Figure 3 presents the distribution of beliefs about the publicity of contribution data for survey respondents from the No-Letter and from the Website treatment groups. Receiving a Website letter did not modify the perception of respondents who reported to be very sure or somewhat sure that the contribution records were confidential. A possible interpretation is that those individuals report this because they interpret the FEC disclosure policy differently: e.g., they may argue that records are confidential because SuperPACs can be used to make veiled contributions, or because

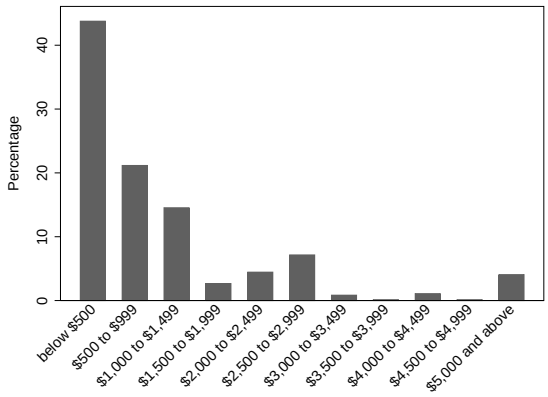
the contribution records are confidential for small donors (i.e., with contributions below the $200 disclosure threshold).

Figure D.1: Knowledge about Contribution Limits and Contribution Norms (Post-Election Mail-In Survey)

a. According to the law, what is the maximum contribution an individual can make to a campaign committee per election?



b. How much do you think a politically engaged individual with an average income should contribute to a presidential campaign per election cycle (every four years)?



Notes: N=3,060 in panel (a) and 2,854 in panel (b). The data corresponds to the responses to our post-election mail-in survey by subjects in the No-Letter group. Panel a presents the histogram of responses to a question about the respondent's knowledge of the maximum contribution level per committee (see question 3 from the questionnaire in Appendix D). Panel b presents the distribution of responses to the survey question which asked recipients to state how much one "should" contribute to a presidential campaign (see question 8 from the questionnaire in Appendix A.7).

E Evidence on the Uses of the FEC Online Search Tool

One potential concern with our findings is that the experimental effects for the conformity channel could be artificial, in the sense that individuals felt social pressure because of our mailing intervention, but would feel this type of pressure in their everyday lives. In this Appendix, we present some descriptive evidence based on the use of the FEC online search tool to argue that this type of social pressure based on campaign contribution records may arise in the course of normal (i.e., not experimentally induced) social interactions. This evidence is based on a proprietary dataset with anonymized records of the browsing history for a sample of millions of Americans from February 1st 2013 to October 31st 2014,[xiv] a period corresponding to the 2014 U.S. Congressional election campaign cycle. This sample includes a small but non-trivial fraction of all visitors to the FEC website and related websites.

A first question is whether the FEC records are being accessed at all. We found that 10% of visitors to the FEC website (www.fec.gov) during that time period used the search tool for individual contributors (www.fec.gov/finance/disclosure/norindsea.shtml). This is a noticeably large share considering that the FEC website provides a number of tools that are not related to searching for individual contributors, and that are used routinely by public officials, politicians, campaign staff members, reporters, and academics. The link to the individual contributor search tool is not even featured on the FEC website's main page: during this time period, a visitor who wanted to reach the search tool on the FEC's website had to click first on "Campaign Finance Disclosure Portal," then on "Search," and finally on "Individual Contributor Search."

The FEC's search tool and the underlying information on contribution records are accessible through other websites as well. According to our browsing data, the number of searches conducted on each of those websites is even larger than the number of searches conducted on the FEC website. For instance, in two of these websites, the number of searches amounted to 133% and 150% of the comparable searches conducted on the FEC's website. While we do not have a reliable way of estimating the total traffic to these websites, our back-of-the-envelope calculations suggest that it is likely that millions of users conduct several million searches each year. Consistent with this, OpenSecrets.org, one of the websites providing a search tool that uses the FEC contribution records, reported nearly 35 million page views to its website in 2012, from more than 5 million unique visitors.[xv]

A second relevant question is what the results from searches of the FEC records are being used for. This is, of course, a much harder question, but we can nonetheless offer some suggestive evidence. First, we can make inferences based on the timing of visits to the FEC online search

[xiv] Unfortunately, we do not have information on the socio-economic characteristics of the individuals included in the sample, so we cannot compare those characteristics with that of the U.S. population or that of contributors.

[xv] Source: https://www.opensecrets.org/about/tour.php, accessed on January 1st, 2015.

tool. If the purpose of accessing contribution records is to learn about candidates, the search tool should be used at a much higher rate in the months preceding an election. Figure E.1 depicts the proportion of visitors to the FEC website that use the online search tool on a monthly basis from February 2013 to October 2014. The percentage of visitors to the FEC's website using the online search tool was roughly stable over the entire period, and it actually decreased as the election (held on November, 2014) drew closer.

We can also infer the purposes behind uses of the FEC online search tool from the types of searches conducted by its users. Our data provides us the criteria used in each search (e.g., searching by name, address, candidate) for one of the alternative websites that offers an online search tool very similar to the FEC's. This search tool allows visitors to search contributors by name, state, ZIP code, employer, and/or candidate. The analysis of these searches reveals that 86% of the searches were conducted with the name of a contributor as the only criterion. Of the remaining 14%, less than half included the name of a candidate, whereas the remaining searches focused on particular ZIP codes or employers. These patterns are also consistent with the conjecture that individuals use the FEC online search tool to seek information about their peers rather than to seek information about the candidates.

While individuals search mainly for specific names, they may not necessarily focus on their peers: they may use these search tools to learn about the contributions of actors, politicians, CEOs and other celebrities. For instance, from time to time U.S. newspapers publish articles about the contributions of celebrities using data from the FEC online search tool. To explore this hypothesis, Figure E.2.a depicts the percentage of the total searches corresponding to names that have been searched for only once, to names that have been searched for twice, and so on. If the primary use for the search tool was to search for celebrities, then most traffic should be directed towards a relatively small number of names (i.e., the celebrities) with hundreds or thousands of searches each. The data strongly reject this hypothesis: the vast majority of the searches are for names that are searched for once or twice in the entire 21-month period (Feb. 2013-Oct. 2014) for which we have internet browsing data.[xvi]

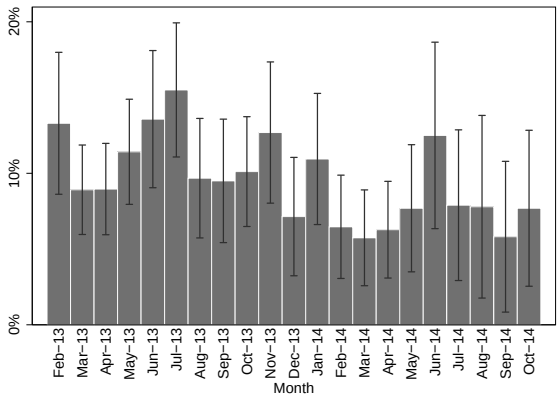
Finally, there would be less scope of social pressure from campaign contribution records if only a few professional users (i.e., campaign staff, fundraisers, etc.) account for the vast majority of searches. We can test this conjecture with our internet browsing data because it has (anonymized but unique) user identifiers. Figure E.2.b depicts the distribution of search activity across users of the FEC search tool. This figure shows the percentage of the total searches that can be accounted for by users conducting 1-10 searches, 11-20 searches, and so on. In the entire 21-month period

[xvi] Individuals may search for the same individual with different terms, i.e. "Angelina Jolie" and "Jolie, Angelina." We did our best to clean up the data to deal with these cases, although of course we have not been able to capture all possible variations. We are confident however that the main patterns are robust to these measurement errors.

before the 2014 congressional election, the majority of users (62.5%) conducted between 1 and 10 searches. If we define professional users as those searching more than 100 times in the entire 21-month period, then this type of user only accounts for 13.7% of the total searches. Professional users, thus, explain a significant share of activity on the FEC online search tool, but they do not account for the majority of the traffic.

In summary, this evidence suggests that millions of Americans (and not only campaign professionals) access the FEC contribution records, possibly to learn about the political affiliation of their peers (neighbors, friends, coworkers, and subordinates) rather than to gather information about candidates.

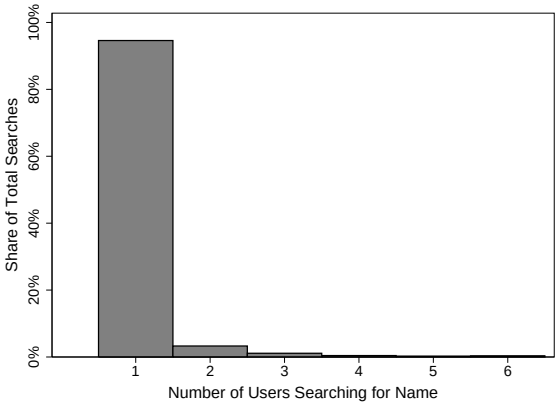
Figure E.1: Visits to the Online FEC Search Tool Over Time, 2013-2014 (Internet Browsing Data)



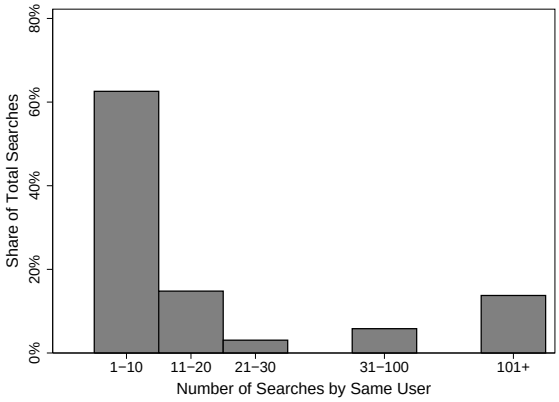
Notes: Share of visitors to the FEC website (www.fec.gov) who conduct at least one search on the FEC online search tool for individual contributors (www.fec.gov/finance/disclosure/norindsea.shtml). The whiskers represent 95% confidence intervals for each monthly average. Source: panel of Internet users.

Figure E.2: Type and Distribution of Searchers using the FEC Online Search Tool (Internet Browsing Data)

a. Number of Searches per Name



b. Number of Searches per User



Notes: Panel (a) presents the percentage of the total searches corresponding to names searched only once, twice, and so no. This data corresponds to all searches conducted in a popular search tool (similar to the FEC's search tool, and based on the same records) where the user completed the "name" field. Panel (b) presents the percentage of the total searches in the FEC online search tool for individual contributors (www.fec.gov/finance/disclosure/norindsea.shtml) that can be accounted for by users conducting 1-10 searches, 11-20 searches, and so on. Source for both figures: panel of Internet users, February-2013 to October-2014.

F Conformity Effects in a Model of Partisan Signaling

F.1 The Model

In this Appendix, we formalize the intuition behind the conformity channel. For that, we present a model where contributions can be used as a signal of the political affiliation of the contributor. This model follows the tradition of a signaling approach to social interactions, as in Bernheim (1994) and in Bénabou and Tirole (2006).

Individuals, indexed by subscript i, can make a discrete contribution denoted by $c_i \in \{-1, 0, 1\}$. $c_i = -1$ means that the individual contributes to the left-wing party, $c_i = 1$ means that the individual contributes to the right-wing party and $c_i = 0$ means that the individual does not contribute to any political party. The discrete nature of contributions is just a convenient simplification to facilitate the tractability of the model. The intuition of the model, however, extends to the case where individuals can make contributions of different amounts. Moreover, even though we are interested specifically in monetary contributions, c_i may also be interpreted as other forms of potentially-observable forms of political participation, such as attending a rally, displaying candidate's yard signs or simply speaking in favor of a candidate.

The individuals belong to reference groups. We will refer to other individuals in i's reference group as i's neighbors. In the empirical analysis, we rely on a geographic proxy for an individual's reference group, but this may represent something more general than just geographic proximity. Reference groups may include family members, friends, acquaintances and co-workers, for instance. A given reference group is comprised by a continuum of agents who differ in a parameter α_i, distributed in the support $[\underline{\alpha}, \overline{\alpha}]$ according to the cumulative distribution function $F_\alpha(\cdot)$, with $\underline{\alpha} < 0$ and $\overline{\alpha} > 0$. The parameter α_i indicates the party supported and the strength of i's political affiliation. Individuals with $\alpha_i < 0$ sympathize with the left-wing party and those with $\alpha_i > 0$ sympathize with the right-wing party. Thus, $S_R = F_\alpha(0)$ and $S_L = 1 - F_\alpha(0)$ are the shares of individuals supporting the left and right parties, respectively. Agent i's utility from contributing to her favorite party is given by $-K + |\alpha_i|$, and her utility from contributing to the opposite party is$-K - |\alpha_i|$. The parameter $K > 0$ represents the fixed cost of contributing, including both pecuniary and non-pecuniary costs. If only these costs were present, individuals with $\alpha_i < -K$ would contribute to the left-wing party, individuals with $\alpha_i > K$ would contribute to the right-wing party, and individuals with $-K < \alpha_i < K$ would refrain from making any contribution.

There are also indirect costs and benefits from making contributions. Each individual belongs to one reference group, and interacts with other members of the same group, which we also refer to as neighbors. In these interactions, some characteristics of the individual are not perfectly observable to her neighbors, such as her affluence, her disposition towards pro-social behavior, her party affiliation and its strength, among many others. The interaction may be more or less

beneficial for the individual depending on what the neighbor perceives about her characteristics. For instance, neighbors may have a kinder or more positive attitude toward individuals that are affluent, that exhibit civic behavior, or that share the neighbor's political preferences. While these characteristics are not directly observable to neighbors, contributions can be observed with certain probability, for instance because neighbors may look up the individual's contribution activity using the FEC website's search tool. Making contributions more visible to neighbors should thus affect the individual's desired contribution level, and this effect should depend on what the neighbor learns from the observed contribution. If contributions signal mostly income or wealth and being perceived as affluent is considered a positive trait, then more visibility should increase contributions. Similarly, if contributions signal pro-social attitudes then more visibility should also increase contributions. However, there are many other more efficient ways to signal affluence and pro-social behavior, for instance buying an expensive car or making named contributions to local charities. It is thus unlikely that individuals use campaign contributions primarily to signal those traits.

When an individual interacts with a neighbor, the utility an individual can expect from this interaction is a function of the coincidence or divergence in political affiliations with the neighbors. Political preferences are not directly observable by others, but contributions may be observed. Whether the individual contributed, and the party contributed to, is visible to i's neighbors with some probability υ, and unobservable with probability $1-\upsilon$.[xvii] Contributions are made prior to the interactions with neighbors.[xviii] When the contribution is observable, a neighbor can infer the individual's political preference - in a probabilistically sense - from the observed contribution (or lack thereof). Let $P_i^j = P^j(c_i, c_{-i})$ be the perceived probability that i sympathizes with party j given i's contribution, c_i, and the vector of everyone else's contributions in the same reference group, c_{-i}. The utility from the interaction with a neighbor of party j is $\delta\left(P_i^j\right)$. The function $\delta(\cdot)$ is monotonically increasing, which means that neighbors treat individuals better when they believe that they support their own political party.

Denote P_i^R the perceived probability that individual i sympathizes with the right-wing party. When her contribution is observable to neighbors, the indirect utility for a right-wing individual is given by:

$$S_R \mu \delta\left(P_i^R\right) + S_L(1-\mu)\delta\left(1-P_i^R\right)$$

This is a weighted average of the expected utilities from interacting with right-wing and left-wing neighbors, where the weights are given by the parameter μ in conjunction with the proportion

[xvii] An alternative interpretation of the probability parameter υ is that the contribution information is always a matter of public record, but each individual is uncertain as to whether her neighbors know about these records or about their publicity, and, if they know about it, whether they would try to access this information.

[xviii] Note that the individual does not know whether her contribution will be observable to others when deciding about her contributions.

of neighbors sympathizing with each party, S_R and S_L. Similarly, the indirect utility for a left-wing individual is given by:

$$S_R (1-\mu)\, \delta\left(P_i^R\right) + S_L \mu \delta\left(1 - P_i^R\right)$$

The parameter $\mu \in \left[\frac{1}{2}, 1\right]$ captures what we denominate political homophily, the tendency of individuals to interact with sympathizers of their own political party. The parameter μ can have one of the following two interpretations (or a combination of the two). First, it may represent differences in the likelihood of meeting a neighbor of each party. Second, it may represent party-based differences in how individuals value the interactions. The case where $\mu = \frac{1}{2}$ corresponds to a situation where individuals are matched with neighbors regardless of their political preferences, while in the case where $\mu > \frac{1}{2}$ each individual is relatively more likely to interact with neighbors supporting their own political party.[xix] Alternatively, $\mu = \frac{1}{2}$ could mean that individuals have the same valuation of interactions with neighbors from either party, while $\mu > \frac{1}{2}$ could indicate that individuals value interactions with own-party neighbors relatively more.

We make two simplifying assumptions to make the model tractable. First, we assume that α_i is uniformly distributed. Second, we assume that $\delta(\cdot)$ is linear. Without any loss of generality, we normalize the intercept of $\delta(\cdot)$ to zero: i.e., $\delta(P) = \gamma \cdot P$. In the signaling equilibrium there will be three groups defined by two thresholds: $\alpha_L^\star \in (\underline{\alpha}, 0)$ and $\alpha_R^\star \in (0, \overline{\alpha})$.[xx] Individuals with $\alpha_i \leq \alpha_L^\star$ will contribute to the left-wing party, individuals with $\alpha_L^\star < \alpha_i < \alpha_R^\star$ will not contribute at all, and individuals with $\alpha_i \geq \alpha_L^\star$ will contribute to the right-wing party. Let Ω_R (Ω_L) denote a right-wing (left-wing) individual's utility from interacting with neighbors when her own contribution is unobservable. The utility for a right-wing individual from contributing to her favorite party is:

$$-K + \alpha_i + \upsilon S_R \mu \gamma + (1-\upsilon)\, \Omega_R$$

The utility for a left-wing individual from contributing to her favorite party is:

$$-K - \alpha_i + \upsilon\, (1 - S_R)\, \mu\gamma + (1-\upsilon)\, \Omega_L$$

The utility of not contributing for a right-wing individual is:

$$\upsilon \left[(S_R - 1 + \mu)\, \gamma \frac{\min\{\alpha_R^\star, \overline{\alpha}\}}{\min\{\alpha_R^\star, \overline{\alpha}\} - \max\{\alpha_L^\star, \underline{\alpha}\}} + (1 - S_R)(1-\mu)\gamma \right] + (1-\upsilon)\, \Omega_R$$

[xix] However, the fact that individuals are more likely to bond with neighbors of the same political party should not be interpreted as an exogenous parameter, i.e., $\mu > \frac{1}{2}$, but instead as part of the indirect costs embedded in $\delta(\cdot)$. That is, revealing oneself as a sympathizer of the opposite party (with respect to the neighbors) is disadvantageous because this results in fewer and/or poorer connections within the reference group.

[xx] Note that we implicitly assume an interior solution.

The utility of not contributing for a left-wing individual is:

$$\upsilon\left[\left(S_R-\mu\right)\gamma\frac{\min\left\{\alpha_R^\star,\overline{\alpha}\right\}}{\min\left\{\alpha_R^\star,\overline{\alpha}\right\}-\max\left\{\alpha_L^\star,\underline{\alpha}\right\}}+\left(1-S_R\right)\mu\gamma\right]+\left(1-\upsilon\right)\Omega_L$$

By construction, $\alpha_R^\star$ is such that a right-wing individual with $\alpha_i=\alpha_R^\star$ is indifferent between contributing to the right-wing party and not contributing at all:

$$\alpha_R^\star=\upsilon\gamma\left(S_R-1+\mu\right)\frac{\min\left\{\alpha_L^\star,\overline{\alpha}\right\}}{\min\left\{\alpha_R^\star,\overline{\alpha}\right\}-\max\left\{\alpha_L^\star,\underline{\alpha}\right\}}+K \tag{6}$$

The analogous expression for a left-wing individuals is:

$$-\alpha_L^\star=\upsilon\gamma\left(S_R-\mu\right)\frac{\min\left\{\alpha_R^\star,\overline{\alpha}\right\}}{\min\left\{\alpha_R^\star,\overline{\alpha}\right\}-\max\left\{\alpha_L^\star,\underline{\alpha}\right\}}+K \tag{7}$$

This system of two equations and two unknowns characterizes the signaling equilibrium.

Denote $\alpha^\star=\{\alpha_L^\star,\alpha_R^\star\}$ and let $\Theta=\left\{\alpha^\star:\ \alpha_L^\star\in\left(\underline{\alpha},-\frac{K}{2}\right),\ \alpha_R^\star\in\left(\frac{K}{2},\overline{\alpha}\right)\ \right\}$. We will focus on equilibria with $\alpha^\star\in\Theta$. The first requirement in Θ is that the solution is interior, i.e., $\underline{\alpha}<\alpha_L^\star<\alpha_R^\star<\overline{\alpha}$.[xxi] The second requirement, $\alpha_L^\star<-\frac{K}{2}<0<\frac{K}{2}<\alpha_R^\star$, basically restricts the analysis to equilibria in which the mass of non-contributors to each party is above the threshold $\frac{K}{2}$. This condition is consistent with the fact that only a small share of individuals contribute to political campaigns. This condition also guarantees that the equilibrium effects described below are of second order and therefore do not override the direct effects of changes in the relevant parameters.[xxii]

Proposition 1. *Given parameter values in a non-empty set* Π*, a signaling equilibrium exists, it is unique and it belongs to* Θ*.*

Proofs of the propositions are provided at the end of this Appendix. While we cannot specify an explicit solution for the model, we can use the implicit function theorem to perform the key comparative statics. The following proposition presents a prediction about the effect of visibility on contributions relevant for the empirical analysis.

Proposition 2. *In any signaling equilibrium from* Θ*, an increase in visibility* (υ) *induces a change in the number of contributors to the majority party that is greater than the change in the number of contributors to the minority party.*

[xxi] It would be straightforward to extend the propositions to the alternative scenario, although the notation would be significantly more complicated. Intuitively, we would need to reproduce the whole analysis for each corner solution.

[xxii] Even though the propositions focus on the more plausible equilibria in Θ (given the fraction of contributors in the actual population), it is straightforward to extend the comparative statics for $\alpha^\star\notin\Theta$ based on the proofs provided here.

If there are more neighbors identified with an individual's party, she will have greater incentives to signal her political preference by making a contribution to that party. In terms of the empirical application presented in this paper, the proposition implies that an exogenous variation in υ should result in a very specific form of heterogeneous effects: the effect of changes in visibility on contributions should be increasing in the share of neighbors supporting the same party as the contributor.

Proposition 3. *In any signaling equilibrium from Θ, an increase in visibility (υ) induces a change in the number of contributors to party j that is positive if $S_j > 1-\mu$, null if $S_j = 1-\mu$, and negative if $S_j < 1-\mu$.*

We should expect changes in υ to have effects of opposite signs on contributions for individuals in two different groups: we should expect a negative effect for those with $S_j < 1-\mu$, and a positive effect for individuals with $S_j > 1-\mu$. For example, if $\mu = \frac{1}{2}$, which denotes a pattern of interactions with neighbors independent of their political preferences, we should expect an exogenous increase in υ to increase contributions for individuals who belong to the majority party in the area, and a reduction in contributions for those identified with the minority party. As a result, an increase in υ in a given reference group should result in one of two scenarios. If $S_j > \mu$ (so that $S_j > 1-\mu$ and $1-S_j < 1-\mu$), greater visibility will increase contributions to the majority party but reduce those to the minority party. Alternatively, if $1-\mu < S_j < \mu$ (so that $S_j > 1-\mu$ and $1-S_j > 1-\mu$), then an increase in visibility will increase contributions to both parties, but (because of Proposition 2) the increase will be greater for the majority party.[xxiii]

Finally, the results also provide a more intuitive interpretation for the condition $\alpha_L^\star < -\frac{K}{2} < 0 < \frac{K}{2} < \alpha_R^\star$. Changes in visibility have both direct and equilibrium effects on contributions. The direct effect is that, holding constant all other agent's contribution patterns, greater visibility makes contributions to a given party either more or less attractive, depending on whether S_j is lower or higher than $1-\mu$. For example, if $\mu = \frac{1}{2}$ then an increase in visibility makes contributions more attractive for the sympathizers of the majority party and less attractive for the sympathizers of the minority party. The equilibrium effect, in turn, results from the fact that other individuals should also react to the change in υ, thereby altering the political composition of the pool of non-contributors. For example, if as a result of a change in υ contributions by individuals identified with the majority party increase, the signal of making no contributions would become more closely associated to being sympathetic to the minority party, thereby changing the value of not making a contribution. When the share of non-contributors is large enough, i.e. $\alpha_L^\star < -\frac{K}{2} < 0 < \frac{K}{2} < \alpha_R^\star$, these equilibrium effects are of second order, so the net effect is dominated by the direct effects.

[xxiii] If we allowed the scenario with $\mu < 1/2$ then a fourth possibility would arise: if $S_j < 1-\mu$ and $1-S_j < 1-\mu$, an increase in visibility decreases contributions to both parties in the reference group, although the fall would be milder for the individuals of the majority party.

However, if the share of non-contributors is very low, then the equilibrium effects may override the direct effects and change the sign of the overall impact of the change in visibility. We only discuss the comparative statics under the more realistic condition where only a small share of individuals make campaign contributions, although it is straightforward to derive predictions under alternative scenarios.

Last, if we define geographic polarization as the difference in contributions to the two parties within a given reference group, the following Corollary is implied by Proposition 2:

Corollary 1. *In any signaling equilibrium from* Θ*, an increase in visibility of contributions* (υ) *will increase geographic polarization.*

F.2 Proof of Propositions

F.2.1 Proof of Proposition 1

From equation (7) we obtain:

$$\alpha_L^{\star 2} + \alpha_L^{\star}\left[K - \alpha_R^{\star}\right] - \left(\upsilon\gamma\left(S_R - \mu\right) + K\right)\alpha_R^{\star} = 0$$

Using the quadratic formula:

$$\alpha_L^{\star} = \frac{\alpha_R^{\star}}{2} - \frac{K}{2} - \sqrt{\left(\frac{\alpha_R^{\star}}{2} - \frac{K}{2}\right)^2 + \left(\upsilon\gamma\left(S_R - \mu\right) + K\right)\alpha_R^{\star}}$$

We only use the left root because the right root cannot have simultaneously $\alpha_L^{\star} < -\frac{K}{2}$ and $\alpha_R^{\star} > 0$. Note that we also need $\left(\upsilon\gamma\left(S_R - \mu\right) + K\right) > 0$, which implies that $\alpha_L^{\star} < 0$. We can replace in equation (6):

$$\alpha_R^{\star} = \upsilon\gamma\left(S_R - 1 + \mu\right)\frac{-\frac{K}{2} + \frac{\alpha_R^{\star}}{2} - \sqrt{\left(\frac{K}{2} - \frac{\alpha_R^{\star}}{2}\right)^2 + \left(\upsilon\gamma\left(S_R - \mu\right) + K\right)\alpha_R^{\star}}}{\alpha_R^{\star} - \left[-\frac{K}{2} + \frac{\alpha_R^{\star}}{2} - \sqrt{\left(\frac{K}{2} - \frac{\alpha_R^{\star}}{2}\right)^2 + \left(\upsilon\gamma\left(S_R - \mu\right) + K\right)\alpha_R^{\star}}\right]} + K$$

and then define:

$$f_R\left(\alpha_R\right) = \upsilon\gamma\left(S_R - 1 + \mu\right)\frac{-\frac{K}{2} + \frac{\alpha_R}{2} - \sqrt{\left(\frac{K}{2} - \frac{\alpha_R}{2}\right)^2 + \left(\upsilon\gamma\left(S_R - \mu\right) + K\right)\alpha_R}}{\alpha_R - \left[-\frac{K}{2} + \frac{\alpha_R}{2} - \sqrt{\left(\frac{K}{2} - \frac{\alpha_R}{2}\right)^2 + \left(\upsilon\gamma\left(S_R - \mu\right) + K\right)\alpha_R}\right]} + K$$

We thus need to prove that a fixed point of $f_R\left(\alpha_R\right)$ exists and is unique in the domain $\alpha_R \in \left(\frac{K}{2}, \overline{\alpha}\right)$. Define $g_R\left(\alpha_R\right) = f_R\left(\alpha_R\right) - \alpha_R$. First we need to prove that $g_R'\left(\alpha_R\right) > 0$. Given that, we would only need to find conditions such that $g_R\left(\frac{K}{2}\right) > 0$ and $g_R\left(\overline{\alpha}\right) < 0$ to prove existence and uniqueness. Starting with $g_R'\left(\alpha_R\right)$:

$$g_R'\left(\alpha_R\right) = \frac{\left[-\frac{K}{2} + \frac{\alpha_R}{2} - \sqrt{\left(\frac{K}{2} - \frac{\alpha_R}{2}\right)^2 + \left(\upsilon\gamma\left(S_R - \mu\right) + K\right)\alpha_R}\right]}{\left(\alpha_R - \left[-\frac{K}{2} + \frac{\alpha_R}{2} - \sqrt{\left(\frac{K}{2} - \frac{\alpha_R}{2}\right)^2 + \left(\upsilon\gamma\left(S_R - \mu\right) + K\right)\alpha_R}\right]\right)^2} - 1$$

To prove that $g_R'\left(\alpha_R\right) < 0$, it is sufficient that:

$$\frac{\alpha_R}{2}-\frac{K}{2}<\sqrt{\left(\frac{\alpha_R}{2}-\frac{K}{2}\right)^2+\left(\upsilon\gamma\left(S_R-\mu\right)+K\right)\alpha_R}$$

If $\frac{\alpha_R}{2}-\frac{K}{2}<0$, this condition is automatically satisfied. If $\frac{\alpha_R}{2}-\frac{K}{2}>0$, we must have $\left(\upsilon\gamma\left(S_R-\mu\right)+K\right)>0$, which we already had to assume. We must then find conditions such that $g_R\left(\frac{K}{2}\right)>0$ and $g_R\left(\overline{\alpha}\right)<0$, where:

$$g_R\left(\frac{K}{2}\right)=\upsilon\gamma\left(S_R-1+\mu\right)\frac{-\frac{1}{4}K-\sqrt{\frac{9}{16}K^2+\upsilon\gamma\left(S_R-\mu\right)\frac{K}{2}}}{\frac{3}{4}K+\sqrt{\frac{9}{16}K^2+\upsilon\gamma\left(S_R-\mu\right)\frac{K}{2}}}+\frac{K}{2}$$

$$g_R\left(\overline{\alpha}\right)=\upsilon\gamma\left(S_R-1+\mu\right)\frac{-\frac{K}{2}+\frac{\overline{\alpha}}{2}-\sqrt{\left(\frac{K}{2}-\frac{\overline{\alpha}}{2}\right)^2+\left(\upsilon\gamma\left(S_R-\mu\right)+K\right)\overline{\alpha}}}{\alpha_R-\left[-\frac{K}{2}+\frac{\overline{\alpha}}{2}-\sqrt{\left(\frac{K}{2}-\frac{\overline{\alpha}}{2}\right)^2+\left(\upsilon\gamma\left(S_R-\mu\right)+K\right)\overline{\alpha}}\right]}+K-\overline{\alpha}$$

We now need to reproduce the entire analysis for α_L: i.e., we need to prove that a fixed point of $f_L\left(\alpha_L\right)$ exists and is unique in the domain $\alpha_L\in\left(\underline{\alpha},-\frac{K}{2}\right)$. From equation (6) we obtain:

$$\alpha_R^\star=\frac{K}{2}+\frac{\alpha_L^\star}{2}+\sqrt{\left(\frac{K}{2}+\frac{\alpha_L^\star}{2}\right)^2-\left(K-\upsilon\gamma\left(S_R-1+\mu\right)\right)\alpha_L^\star}$$

In this expression, we need to assume that $\left(K-\upsilon\gamma\left(S_R-1+\mu\right)\right)>0$. From the following:

$$f_L\left(\alpha_L\right)=-\upsilon\gamma\left(S_R-\mu\right)\frac{\frac{K}{2}+\frac{\alpha_L^\star}{2}+\sqrt{\left(\frac{K}{2}+\frac{\alpha_L^\star}{2}\right)^2-\left(K-\upsilon\gamma\left(S_R-1+\mu\right)\right)\alpha_L^\star}}{\left[\frac{K}{2}+\frac{\alpha_L^\star}{2}+\sqrt{\left(\frac{K}{2}+\frac{\alpha_L^\star}{2}\right)^2-\left(K-\upsilon\gamma\left(S_R-1+\mu\right)\right)\alpha_L^\star}\right]-\alpha_L^\star}-K$$

we can proceed in a similar manner than for $f_R\left(\alpha_R\right)$, since $\left(K-\upsilon\gamma\left(S_R-1+\mu\right)\right)>0$, $g_L'\left(\alpha_L\right)<0$. To sum up, if the parameter values belong to the following set then an equilibrium exists, it is unique and it belongs to Θ:

$$\Pi=\left\{\begin{array}{c}\left\{K,\mu,\underline{\alpha},\overline{\alpha},\upsilon,\gamma\right\}:\ g_R\left(\frac{K}{2}\right)>0,\ g_R\left(\overline{\alpha}\right)<0,\ g_R\left(\underline{\alpha}\right)>0,\ g_L\left(-\frac{K}{2}\right)<0,\\ K>\max\left\{-\upsilon\gamma\left(S_R-\mu\right),\upsilon\gamma\left(S_R-1+\mu\right)\right\}\end{array}\right\}$$

Finally, it is trivial to prove that Π is non-empty by means of an example.

F.2.2 Proof of Proposition 2

Denote $C_R = \frac{\overline{\alpha}-\alpha_R^\star}{\overline{\alpha}-\underline{\alpha}}$ as the mass of individuals contributing to the right-wing party and $C_L = \frac{\alpha_L^\star-\underline{\alpha}}{\overline{\alpha}-\underline{\alpha}}$ as the mass of individuals contributing to the left-wing party. It follows that:

$$\frac{dC_R}{d\upsilon} - \frac{dC_L}{d\upsilon} = \frac{1}{\overline{\alpha}-\underline{\alpha}}\left[-\frac{d\alpha_R^\star}{d\upsilon} - \frac{d\alpha_L^\star}{d\upsilon}\right]$$

We need to prove that $S_R > \frac{1}{2}$ implies that $\frac{dC_R}{d\upsilon} - \frac{dC_L}{d\upsilon} > 0$. To establish this, we need to obtain expressions for $\frac{d\alpha_R^\star}{d\upsilon}$ and $\frac{d\alpha_L^\star}{d\upsilon}$. We will calculate those derivatives using the implicit function theorem. We start by defining:

$$F(\upsilon, \alpha_R^\star, \alpha_L^\star) = \begin{bmatrix} \alpha_R^\star - \upsilon\gamma\left(S_R - 1 + \mu\right)\frac{\alpha_L^\star}{\alpha_R^\star-\alpha_L^\star} - K \\ -\alpha_L^\star - \upsilon\gamma\left(S_R - \mu\right)\frac{\alpha_R^\star}{\alpha_R^\star-\alpha_L^\star} - K \end{bmatrix}$$

$$H = \begin{bmatrix} \frac{dF_1}{d\alpha_R^\star} & \frac{dF_1}{d\alpha_L^\star} \\ \frac{dF_2}{d\alpha_R^\star} & \frac{dF_2}{d\alpha_L^\star} \end{bmatrix} = \begin{bmatrix} 1 + \upsilon\gamma\left(S_R - 1 + \mu\right)\frac{\alpha_L^\star}{\left(\alpha_R^\star-\alpha_L^\star\right)^2} & -\upsilon\gamma\left(S_R - 1 + \mu\right)\frac{\alpha_R^\star}{\left(\alpha_R^\star-\alpha_L^\star\right)^2} \\ \upsilon\gamma\left(S_R - \mu\right)\frac{\alpha_L^\star}{\left(\alpha_R^\star-\alpha_L^\star\right)^2} & -1 - \upsilon\gamma\left(S_R - \mu\right)\frac{\alpha_R^\star}{\left(\alpha_R^\star-\alpha_L^\star\right)^2} \end{bmatrix}$$

$$M^\upsilon_{\alpha_R^\star} = \begin{bmatrix} \frac{dF_1}{d\upsilon} & \frac{dF_1}{d\alpha_L^\star} \\ \frac{dF_2}{d\upsilon} & \frac{dF_2}{d\alpha_L^\star} \end{bmatrix} = \begin{bmatrix} -\gamma\left(S_R - 1 + \mu\right)\frac{\alpha_L^\star}{\alpha_R^\star-\alpha_L^\star} & -\upsilon\gamma\left(S_R - 1 + \mu\right)\frac{\alpha_R^\star}{\left(\alpha_R^\star-\alpha_L^\star\right)^2} \\ -\gamma\left(S_R - \mu\right)\frac{\alpha_R^\star}{\alpha_R^\star-\alpha_L^\star} & -1 - \upsilon\gamma\left(S_R - \mu\right)\frac{\alpha_R^\star}{\left(\alpha_R^\star-\alpha_L^\star\right)^2} \end{bmatrix}$$

$$M^\upsilon_{\alpha_L^\star} = \begin{bmatrix} \frac{dF_1}{d\alpha_L^\star} & \frac{dF_1}{d\upsilon} \\ \frac{dF_2}{d\alpha_L^\star} & \frac{dF_2}{d\upsilon} \end{bmatrix} = \begin{bmatrix} 1 + \upsilon\gamma\left(S_R - 1 + \mu\right)\frac{\alpha_L^\star}{\left(\alpha_R^\star-\alpha_L^\star\right)^2} & -\gamma\left(S_R - 1 + \mu\right)\frac{\alpha_L^\star}{\alpha_R^\star-\alpha_L^\star} \\ \upsilon\gamma\left(S_R - \mu\right)\frac{\alpha_L^\star}{\left(\alpha_R^\star-\alpha_L^\star\right)^2} & -\gamma\left(S_R - \mu\right)\frac{\alpha_R^\star}{\alpha_R^\star-\alpha_L^\star} \end{bmatrix}$$

By the implicit function theorem, we know that:

$$\frac{d\alpha_R^\star}{d\upsilon} = -\frac{det\left(M^\upsilon_{\alpha_R^\star}\right)}{det\left(H\right)} = \gamma\left(S_R - (1-\mu)\right)\frac{\alpha_L^\star + \frac{K}{2}}{-\left(\alpha_L^\star + \frac{K}{2}\right) + \left(\alpha_R^\star - \frac{K}{2}\right)}$$

Then, for $\frac{d\alpha_L^\star}{d\upsilon}$:

$$\frac{d\alpha_L^\star}{d\upsilon} = -\frac{det\left(M^\upsilon_{\alpha_L^\star}\right)}{det\left(H\right)} = \gamma\left(S_L - (1-\mu)\right)\frac{\alpha_R^\star - \frac{K}{2}}{-\left(\alpha_L^\star + \frac{K}{2}\right) + \left(\alpha_R^\star - \frac{K}{2}\right)}$$

Finally, we can replace back in $\frac{dC_R}{d\upsilon} - \frac{dC_L}{d\upsilon}$:

$$\frac{dC_R}{d\upsilon} - \frac{dC_L}{d\upsilon} = \frac{1}{\overline{\alpha} - \underline{\alpha}}\gamma \left[\frac{\left[-\left(\alpha_L^\star + \frac{K}{2}\right) + \left(\alpha_R^\star - \frac{K}{2}\right)\right]\left(S_R - \frac{1}{2}\right) - \left(\mu - \frac{1}{2}\right)\left[\alpha_L^\star + \alpha_R^\star\right]}{-\left(\alpha_L^\star + \frac{K}{2}\right) + \left(\alpha_R^\star - \frac{K}{2}\right)} \right]$$

Combining $F_1(\cdot) = 0$ and $F_2(\cdot) = 0$, we know that:

$$\alpha_R^\star + \alpha_L^\star = \upsilon\gamma\left(S_R - \frac{1}{2} + \left(\mu - \frac{1}{2}\right)\right)\frac{\alpha_L^\star}{\alpha_R^\star - \alpha_L^\star} - \upsilon\gamma\left(S_R - \frac{1}{2} + \left(\frac{1}{2} - \mu\right)\right)\frac{\alpha_R^\star}{\alpha_R^\star - \alpha_L^\star}$$

Plugging this expression in the previous equation, we obtain:

$$\frac{dC_R}{d\upsilon} - \frac{dC_L}{d\upsilon} = \frac{1}{\overline{\alpha} - \underline{\alpha}}\gamma \left[\frac{\left[-\left(\alpha_L^\star + \frac{K}{2}\right) + \left(\alpha_R^\star - \frac{K}{2}\right) + \left(\mu - \frac{1}{2}\right)\upsilon\gamma\right]\left(S_R - \frac{1}{2}\right) + \upsilon\gamma\left(\mu - \frac{1}{2}\right)^2}{-\left(\alpha_L^\star + \frac{K}{2}\right) + \left(\alpha_R^\star - \frac{K}{2}\right)} \right]$$

If $\mu \geq \frac{1}{2}$, then $S_R > \frac{1}{2}$ implies $\frac{dC_R}{d\upsilon} - \frac{dC_L}{d\upsilon} > 0$, which is exactly what we needed to prove.

F.2.3 Proof of Proposition 3

Recall the values of $\frac{dC_R}{d\upsilon}$ and $\frac{dC_L}{d\upsilon}$ from Proof of Proposition 2:

$$\frac{dC_R}{d\upsilon} = -\frac{1}{\overline{\alpha} - \underline{\alpha}}\frac{d\alpha_R^\star}{d\upsilon} = \gamma\frac{S_R - (1-\mu)}{\overline{\alpha} - \underline{\alpha}}\frac{-\left(\alpha_L^\star + \frac{K}{2}\right)}{-\left(\alpha_L^\star + \frac{K}{2}\right) + \left(\alpha_R^\star - \frac{K}{2}\right)}$$

$$\frac{dC_L}{d\upsilon} = \frac{1}{\overline{\alpha} - \underline{\alpha}}\frac{d\alpha_L^\star}{d\upsilon} = \gamma\frac{S_L - (1-\mu)}{\overline{\alpha} - \underline{\alpha}}\frac{\alpha_R^\star - \frac{K}{2}}{-\left(\alpha_L^\star + \frac{K}{2}\right) + \left(\alpha_R^\star - \frac{K}{2}\right)}$$

Since $\alpha^\star \in \Theta$, we have $-\left(\alpha_L^\star + \frac{K}{2}\right) > 0$, $\alpha_R^\star - \frac{K}{2} > 0$ and $-\left(\alpha_L^\star + \frac{K}{2}\right) + \left(\alpha_R^\star - \frac{K}{2}\right) > 0$. It is straightforward to verify that the sign of $\frac{dC_j}{d\upsilon}$ is positive if $S_j > 1 - \mu$, null if $S_j = 1 - \mu$, and negative if $S_j < 1 - \mu$.